BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

The Acute Effect of Breaking up Prolonged Sitting on Cognitive and Cerebrovascular Function: A Systematic Review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-050458
Article Type:	Original research
Date Submitted by the Author:	20-Feb-2021
Complete List of Authors:	Chueh, Ting-Yu; National Taiwan Normal University, Physical Education Chen, Yung-Chih; National Taiwan Normal University, Department of Physical Education Hung , Tsung-Min ; National Taiwan Normal University, Department of Physical Education
Keywords:	MENTAL HEALTH, PUBLIC HEALTH, NEUROPHYSIOLOGY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Breaking sitting and Cognition

Title: The Acute Effect of Breaking up Prolonged Sitting on Cognitive and Cerebrovascular Function: A Systematic Review

Running head: Breaking sitting and Cognition

Word count: 4463

Authors:
Ting-Yu Chueh[1], Yung-Chih Chen[1], Tsung-Min Hung[1, 2]

Affiliations:
[1] Department of Physical Education, National Taiwan Normal University, Taipei, Taiwan
[2] Institute for Research Excellence and Learning Science, National Taiwan Normal University, Taipei, Taiwan

Correspondence:
Yung-Chih Chen, Ph.D.
Department of Physical Education, National Taiwan Normal University, Taipei, Taiwan
No.162, Sec. 1, Heping E. Rd., Da'an Dist., Taipei, Taiwan (R.O.C)
E-mail: yc.chen@ntnu.edu.tw

Tsung-Min Hung, Ph.D., FNAK, FISSP
Department of Physical Education & Institute for Research Excellence and Learning Science, National Taiwan Normal University
No.162, Sec. 1, Heping E. Rd., Da'an Dist., Taipei, Taiwan (R.O.C)
Tel: 886-2-7734-3202; Fax: 886-2-2363-4240
E-mail: ernesthungkimo@yahoo.tw

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and Cognition

Abstract

Objectives: The aim of this study was to review the current evidence on the acute effects on cognitive and cerebrovascular function of breaking prolonged periods of sitting with intermittent physical activity (PA) in healthy populations.

Design: Systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines.

Methods: The four electronic databases including PubMed, Scopus, MEDLINE, and ProQuest were searched for eligible studies. Studies were included if they investigated the acute effects of taking regular PA breaks from sitting on cognitive and cerebrovascular function in healthy populations.

Results: Nine studies were eligible for inclusion in this review. Results from cognitive function, three of the seven studies found positive effects of sitting interrupted by either a) 3 minutes of relatively high-intensity (6 km/h) walking every 30 minutes on attention and inhibitory control in young adults; b) hourly sitting breaks with progressive periods (10–30 minutes) of very light-intensity cycling/walking on attention, working memory and cognitive flexibility in adults with obesity; and c) one bout of continuous exercise with subsequent 3 minutes light-intensity (3.2 km/h) walking breaks every 30 minutes on working memory in older adults with overweight. Two out of three studies on cerebrovascular function found that taking short duration periods (2–3 minutes) of light-intensity breaks from sitting resulted in greater middle cerebral artery blood velocity (MCAv) or temporal MCAv across day compared to prolonged sitting regardless of including an initial bout of continuous exercise.

Conclusion: The small number of available studies provided preliminary results that interrupting periods of sitting might beneficially affect brain function in healthy adults. Given that the studies differed in terms of the nature of their interventions, the optimal pattern of sitting breaks for cognitive health needs to be investigated further.

Trial registration number PROSPERO CRD42020146536.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and Cognition

Key words: Sedentary behavior; Cognition; Physical activity; Breaking sitting; Brain

Strengths and limitations of this study

- ✓ This is the first study to systematically review the acute effects of taking regular activity breaks from sitting on cognitive and cerebrovascular function in healthy individuals.
- ✓ This study was carried out with methodological rigour, including explicit eligibility criteria, a broad extensive database search, studies reviewed by experienced authors working in pairs and independently and evaluation of the risk of bias.
- ✓ The current review provides the potential underlying mechanisms behind the acute effect of taking physical activity breaks from sitting on cognitive and cerebrovascular function.
- ✓ A meta-analysis was not possible to conduct due to the methodological heterogeneity.
- ✓ Solid conclusions are limited by the small number of the available studies.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and Cognition

Introduction

Better cognitive function allows for optimizing daily behaviors through several domains including attention, executive function, and memory engagement [1]. Impaired cognitive function has been associated with adverse change in cerebrovascular function that describes the mechanism of regulating and maintaining constant cerebral perfusion of cerebral blood flow (CBF) to provide sufficient energy to the brain [2]. Studies have shown that temporary reductions in CBF are linked to impaired cognitive function [3], whilst in the longer term possibly result in several neurodegenerative diseases such as dementia [4,5]. Therefore, it is important to maintain and/or enhance cognitive and cerebrovascular function in daily life.

Adults typically spend the vast majority of time (average 8–10 hours) of their waking day in sedentary behavior such as sitting and/or reclining [6,7]. Recent studies have indicated that increased sedentary behavior was associated with lower cognitive performance [8-10]. Furthermore, studies also have demonstrated that prolonged sitting transiently deteriorated cognitive performances (i.e., increases problem solving and attention task errors) [11,12] and decreased CBF [13]. It has been well established that cognitive performances are enhanced following the cession of 20 to 30 minutes continuous/structured exercise [14,15], but individuals, who engage in such exercise, may be largely sedentary throughout the rest of the day [16]. Furthermore, it might not be feasible for some populations (e.g., those who are overweight/obese) to take part in continuous/structured exercise on a daily basis. Accordingly, it would be very useful to understand the optimal mix of physical activity (PA) and sedentary behavior in the course of each day.

Taking regular activity breaks from sitting, especially interrupting prolonged periods of sitting with frequent short bouts of PA, could be a practical approach to reduce the negative impact of sedentary behavior on cognitive and cerebrovascular function. While recent studies have investigated the effects of taking sitting breaks on cognition, these studies have not been subject to a systematic review, which would provide a high level of confidence in the current

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

findings. Furthermore, understanding the physiological mechanisms behind the acute effect of taking PA breaks from sitting is important to facilitate the design of successful approaches. Thus, the aim of this paper was to (1) systematically review studies investigating the acute effect of breaking up prolonged periods of sitting on cognitive and cerebrovascular function compared with uninterrupted sitting; (2) provide a methodological assessment of all included studies; (3) discuss the potential underlying physiological mechanisms for any effects found and provide suggestions for future studies.

Methods

This review was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines [17], and is registered at the International Prospective Register of Systematic Reviews (PROSPERO) (identification code: CRD42020146536).

Eligibility criteria

Inclusion of articles were determined by the application of the PICOS procedures (Population, Intervention, Comparison, Outcome, and Study design) [17]. The inclusion criteria were as follows: (P) Participants: participants were healthy without any cardiovascular related disease, history of brain injury, psychiatric or neurological disorders; (I) Intervention: interventions were of an acute nature, involving interrupting prolonged periods of sitting with regular pattern of PA at predefined interval (e.g., every 30 minutes); (C) Comparator: studies included a non-treatment control groups (i.e., prolonged sitting); (O) Outcome: studies were required to test either cognitive performance and/or cerebrovascular response; and (S) Study design: randomized controlled or non-randomized controlled trials. Furthermore, studies were excluded if they involved the following: (1) only a single bout of PA in the intervention

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

condition; (2) PA related parameters not explicitly stated, such as type, intensity, frequency, and duration; (3) not published in an English language peer-reviewed journal.

Information sources and Search Strategy

Four electronic databases, PubMed, Scopus, MEDLINE, and ProQuest were searched. While no starting date of search was specified, the initial search conducted on the August of 5th, 2019 and final search was completed on the September of 20th, 2020. Key words were discussed in the research team to maximally identify the potential articles. Key words were as follows: (*"Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting"*) *AND* (*cognitive OR cognition OR executive OR brain OR cerebral*).

Inclusion or exclusion of articles were decided with reference to the PICOS criteria, with the screening and selection of studies being completed by two of the authors (TYC and YCC). First, titles and abstracts were independently assessed by these two authors and initially coded as 'yes', 'no' or 'maybe'. The same two authors then reviewed the full text of the 'yes' and 'maybe' studies and if they were not able to agree as to whether it should be included, the matter referred to the third reviewer (TMH) for determination. Reference lists for all included papers were then searched to check for potential additional relevant studies. **Figure 1** provides an overview of the selection process.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and Cognition

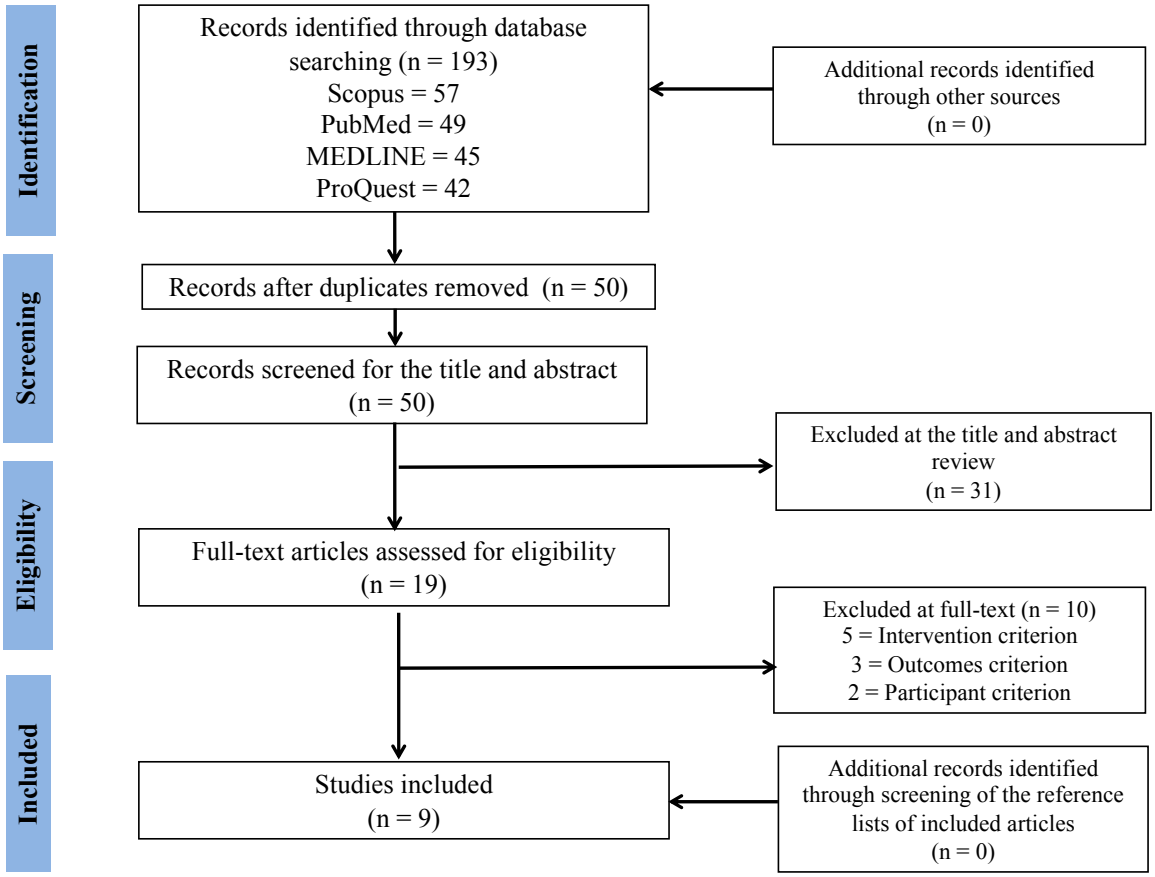


Figure 1. Flow diagram for identification, screening, eligibility, and inclusion of studies.

Data Extraction

Data extraction were conducted by two authors (TYC and YCC). The final studies were examined thoroughly, and the following data were extracted: (1) first author's name, publication year, and country of data collection; (2) participants' characteristics (i.e., sample size, age, weight status, duration of exercise and daily sedentary behavior) and (3) study design; (4) details about intervention protocol; (5) outcomes measurement. **Table 1** summarizes the characteristics of included studies.

Synthesis of Results

The results in this review were analyzed by through a process of narrative synthesis. Included studies were analysed by one reviewer (TYC) and then verified by a second reviewer

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

(YCC). Outcome measures were categorized as pertaining to either 1) cognitive performance (e.g., reaction times, accuracy, etc.); or 2) cerebrovascular response that related to cerebral blood flow (e.g., middle cerebral artery blood velocity or prefrontal cortex perfusion). All the necessary information was obtained via studies published on-line, therefore; no authors were contacted for the relevant information. A p-value of < 0.05, presented in the original studies, was used across studies to determine significant effects of breaking sitting with PA on cognitive and cerebrovascular function. These were converted to effect sizes (standardized mean difference) and entered into the results column of **Table 1**. Effect sizes were interpreted according to the following: small ($d = 0.2$), medium ($d = 0.5$) and large ($d = 0.8$) [18].

Quality Assessment

The quality of the included studies in the systematic review was assessed against the PEDro scale [19]. The PEDro scale includes criteria on allocation bias (randomization, concealment of allocation), performance bias (blinding of participants and personnel) and detection bias (blinding of outcome assessment). The remaining criteria concern: eligibility criteria, baseline comparability, retention rate, intention-to-treat, between-group statistical comparisons for at least one key outcome, point measures and measures of variability. When available, the PEDro score was obtained from the PEDro database. In two studies, the PEDro score were obtained from the PEDro database [20,21]. If a score was not available from the database, it was calculated by two review authors (TYC, and YCC) independently, both of whom had undergone PEDro training. Differences were resolved by discussion and adjudicated by a third author (TMH) as required.

Results

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

In total, nine studies met all the inclusion criteria and are included in this review with the final search taking place on September of 20th, 2020. **Figure 1** illustrates the selection process using a PRISMA flow diagram [17].

Characteristics of Included Studies

All nine included studies were randomized crossover trials, and there was a total of 97 male and 100 female participants aged between 18 to 80 years old. The body mass index (BMI) of participants across all studies was between 18.5 to 44 kg/m^2. Most of participants (54%) were overweight or obese (BMI > 25 kg/m^2) [20-23]. Most of participants (75%) did not report being actively engaged in regular exercise, defined as engaging in moderate-to-vigorous intensity physical activity (MVPA) for at least 150 minutes per week. Sedentary behavior, across all studies, was reported as being between 5 to 9 hours. **Table 1** provides a summary of the study designs and participant characteristics.

Intervention protocols

Regarding the intensity of PA during the breaks, six out of the nine studies stated that the exercise was of light-intensity (walking on a treadmill at 1.6–3.6 km/h or cycling at 20 watts with 25–30 rpm) [20-25], two studies reported that the break consisted of moderate-intensity PA (walking on a treadmill at ~ 6 km/h) [26], and one study implemented 10 calf raises exercise at a rate of 0.33 Hz (20 per minutes) [27].

With regard to duration and frequency of sitting breaks, three studies took 3 minutes PA breaks from sitting every 30 minutes for a total daily exercise duration of 27 [26], 30 [23], and 51 minutes [25]. One study had every 60 minutes of sitting interrupted by 5 minutes of PA with a daily total of 30 minutes PA [28], one study had every 60 minutes of sitting interrupted by progressively longer duration of either walking or cycling (i.e., 10, 15, 20, and 30 minutes) with a daily total of 150 minutes very low-intensity PA [22], and one study employed two

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

intervention arms that varied in terms of both frequency and duration of breaks, with a total of 16 minutes PA (2 minutes walking for every 30 minutes of sitting *versus* 8 minutes of walking for every 120 minutes sitting) [24]. One study instructed participants to performed 10 calf raises exercise within 30 seconds every 5 or 10 minutes with total PA duration being around 9 minutes (17 sets of calf exercise) [27]. In the remaining two studies, there was an initial 30-minute bout of moderate-intensity walking and thereafter sitting was interrupted every 30 minutes by 3 minutes of PA so that the total duration of PA for these two studies was 69 minutes [20,21]. **Table 1** summarises the intervention characteristics for all included studies.

Cognitive function measurements and results

Seven studies examined the acute effects of breaking sitting on cognitive performance [21-23,25-28], with all studies focused on executive function (EF), three studies looked at attention or processing speed [21,22,26], two targeted memory [22], and one study employed a visual learning task [21]. Of these studies, both progressively walking and cycling for longer periods of light-intensity every 60 minutes resulted in greater attention, working memory, and cognitive flexibility compared to sitting with small to large effects (ES = 0.42 to 1.1) [22]. Interestingly, cycling resulted in better attention than walking with a medium effect (ES = 0.63). Taking 3 minutes of moderate-intensity PA breaks every 30 minutes resulted in larger effects on attention (ES = 0.89) and inhibitory control (ES = 0.74) [26] while there were no significant effects of taking 3 minutes of light-intensity walking breaks every 30 minutes and 5 minutes of moderate-intensity walking breaks every 60 minutes on any cognitive performance measures [23,25]. Three minutes of light-intensity PA breaks every 30 minutes with an initial bout of continuous exercise had a beneficial impact on working memory (ES = 0.25) [21]. One study found adverse effects of intermittent 10 calf raise exercises on inhibitory control (ES = -0.85) [27].

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Cerebrovascular function measurements and results

Of the three studies that assessed the effects of taking PA breaks from sitting on cerebrovascular response, two measured middle cerebral artery blood velocity (MCAv) via transcranial Doppler (TCD) [20,24], and one assessed prefrontal cortex perfusion via the near infrared spectroscopy (NIRS) [27]. Two minutes of light-intensity walking breaks every 30 minutes resulted in higher mean seated MCAv in young adults compared to both 8 minutes light-intensity walking breaks every 120 minutes (ES = 0.79) and prolonged sitting (ES = 1.16), with there being no significant difference between later conditions (ES = 0.18) [24]. A positive effect on temporal MCAv was observed after taking 3 minutes light-intensity PA between sitting sessions after an initial a bout of continuous exercise [20]. One study found adverse effects of intermittent 10 calf raise exercises on prefrontal cortex perfusion (ES = -0.97) [27].

Methodological Quality

The included studies had a mean score of 4.2 on the PEDro scale, which is described as 'fair'. All studies were reported to have randomly allocated participants in the trials, however only two studies reported performing a concealed allocation [20,21], which might suggest the presence of allocation biases in the other seven studies. None mentioned the blinding of participants and researchers because it is impossible to blind an intervention which requires the participants to do activities. All studies provided a key outcome measure for more than 85% of their participants. All studies adopted a crossover trials design, which meant that all participants received all designed treatments. Nevertheless, none of the studies explicitly stated whether at least one main outcome was included in the statistical analysis (intention-to-treat analysis) of all participants regardless of any subsequent withdrawal from treatment. As such this might indicate the presence of biased comparisons among the treatment arms. Finally, all studies provided statistical comparisons and valid measures for at least one key outcome measure of

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

interest such as cognitive performance [21-23,25-28] or cerebrovascular response measurements [20,24,27]. Detailed results from the PEDro assessment are presented in **Table 2**.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Discussion

The purpose of this study was to review the results of studies which investigated whether breaking prolonged periods of sitting would result in better cognitive or cerebrovascular function. A total of nine studies were identified which investigated the acute effects of such interruptions in healthy individuals. All studies were published recently (between 2016 and 2020). To date, the small number of available studies (3 out of 7 studies of cognitive function; 2 out of 3 studies of cerebrovascular function) provide preliminary evidence that taking regular activity breaks from sitting offers short-term benefits for cognitive and cerebrovascular functioning in healthy adults.

The acute effects of breaking sitting on cognitive performances

This review suggests that the effect of breaking up sitting with PA on cognitive performance is moderated by the nature of the PA. For example, breaking prolonged periods of sitting every 30 minutes with 3 minutes of moderate-intensity walking (6 km/h) resulted in better attention and inhibitory control in young female adults [26]. However, the same protocol (3 minutes PA every 30 minutes) but involving light-intensity walking speed (3.2 km/h) produced no effect on episodic memory, inhibitory control, or WM in overweight/obese middle-aged adults [23], or on attention and WM in young adults [25]. Furthermore, a meta-analysis demonstrated that the positive acute effect of exercise on cognition is seen with moderate-intensity exercise, but the effect is diminished when with very light-intensity exercise [29]. One potential explanation might be that greater exercise intensity is needed to induce sufficient levels of norepinephrine (NE) to facilitate cognitive performance [30]. Thus, the intensity of PA during the breaks may be one of the factors determining the effectiveness of the interruption.

However, taking physical activity break with higher intensity is not always beneficial if the frequency of such breaks is low. For example, another study of non-obese adults investigated the effects of a similar walking speed (~5.7 km/h) but less frequent (hourly) but

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

slightly longer exercise periods (5 minutes), and no effect was observed in measures of inhibitory control, or cognitive flexibility [28]. This implies that the benefits of taking relatively high intensity PA breaks from sitting might be reduced by having infrequent breaks. Interestingly, taking low intensity PA break might still benefit cognition if the PA duration is long enough. As indicated by [22] both very low intensity cycling (work rate = 20 watts, cadence = 25–30 rpm) and walking (1.6 km/h) resulted in better levels of attention (the detection test), cognitive flexibility (set shifting test), and WM (one back test) compared to prolonged sitting. Despite the low intensity of PA break utilized, the longer duration resulted in a greater amount of exercise (150 minutes in total), which might have had a compensating effect. This finding implies that volume of exercise should be considered and carries important implication for some individuals who are unable to engage in moderate to vigorous intensity exercise. Collectively, these results suggest that the effects of taking breaks from sitting on cognitive function should take into account the intensity, frequency, and duration of the activity during breaks. It would be particularly useful if future studies were able to determine the minimum number and duration of sitting breaks involving low intensity PA that result in improved cognition.

The study involving a morning bout of 30 minutes of moderate-intensity exercise with and without subsequent light-intensity PA breaks every 30 minutes on cognitive performance in obese/overweight older adults provides another fruitful line of research [21]. A single bout of moderate-intensity PA before engagement in prolonged sitting showed better EF, but not WM relative to the sitting condition, whereas post-exercise sitting interrupted by several additional bouts of light-intensity PA led to improved WM, but not EF [21]. These results suggest that a single bout of continuous moderate-intensity exercise with or without subsequent PA breaks provide benefits in different cognitive domains. These divergent effects may be attributed to individual differences in the level of aerobic fitness, and the timing and order of the administration of cognitive tests following the cessation of exercise. It is reasonable to assume

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

that participants in that study had lower aerobic fitness based on the provided demographic data (old age, sedentary lifestyle, and obese/overweight status). In this study, the test of EF was administered almost immediately (approximately 2 minutes) after the 3-minute light-intensity PA breaks, followed by the WM test and other cognitive assessments. Previous meta-analytic review indicated that lighter intensity PA had its largest effect on cognitive performance immediately after PA, whereas cognitive enhancements produced by higher intensity activities were greatest after a delay of ~10 min [29]. However, an individual's level of aerobic fitness may moderate the physiological demands and/or perception of the physical activity stimulus [31]. Chang, et al. [29] observed that aerobic fitness was a moderating variable when cognition was assessed immediately after the cessation of a bout of PA. Thus, the beneficial effects of a single bout of continuous moderate-intensity exercise with subsequent light-intensity PA breaks on EF in this study might be diminished due to the lower aerobic fitness level of participants and small time interval between post-exercise and EF assessment. Moreover, when the aim is to maintain WM, it would appear that it is necessary to avoid prolonged sitting even after an exercise session.

Of note, one study found adverse effects of frequently calf raise exercises on inhibitory control in young adults [27]. The impaired inhibitory control performance was accompanied by a decreased prefrontal cortex perfusion as measured by the NIRS. This finding provided further support for the importance of the prefrontal cortex engagement during inhibitory control performance. The negative impact of acute exercise on executive function has been observed only after a bout of high intensity exercise [29]. We are unaware of any studies reported a negative effect following a short bout of low-intensity exercise such as a calf raise exercise used in this study. The reason for this result is unknown and this result should be replicated in future studies.

Taken together, the literature reviewed suggests that the effects on cognitive performance of breaking periods of sitting with PA needs to concurrently consider the effects of frequency, intensity, and duration of the PA engaged in during the break. The current evidence suggested

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

that either a) short, frequent (at least every 30 minutes) breaks involving moderate intensity PA (walking at least 6 km/h); b) hourly longer breaks involving even very low-intensity PA; and c) a single bout of 30 minutes continuous moderate-intensity exercise with subsequent short PA breaks, all might be beneficial to cognitive performance (e.g., attention, EF, or memory-related tasks). However, the heterogeneity of study protocols and the small number of studies makes direct comparisons difficult.

The acute effects of breaking sitting on cerebrovascular responses

Two out of three studies of cerebrovascular responses showed that interrupted sitting reduced the negative effect of prolong sitting on cerebral blood flow (CBF) [20,24], which might potentially prevent the risk of dementia [32,33]. One study found that prolonged sitting broken with 2 minutes of light-intensity walking every 30 minutes resulted in higher mean middle cerebral artery velocities (MCAv) in young adults compared to both prolonged sitting with 8 minutes light-intensity walking breaks every 120 minutes, and prolonged sitting [24]. In this study, uninterrupted sitting induced a decline in MCAv of 1.4–3.2 cm/s from morning (7 to 9 AM) to midday (11 AM to 1 PM). It has been suggested that the reduction of MCAv observed following 4-hour of uninterrupted sitting equates to 2 to 4 years (0.76 cm/s per year) of age-related decline [32] and whereas frequent light intensity walking breaks may be able to offset this. A previous study found that the lowest CBF occurred at midday (around 12 PM) [13] due to the circadian rhythm [34]. This suggests that the frequency of taking breaks from sitting might be an important factor in the regulation of cerebrovascular response between morning to midday in adults. For older, overweight or obese individuals' MCAv declined across a 8-hour period of sitting, but if the sitting was preceded with a morning bout of 30 minutes of moderate-intensity exercise this decline was eliminated regardless of whether the sitting was interrupted (every 30 minutes for 3 minutes) [20]. Surprisingly, MCAv was higher for those who engaged in the morning bout of exercise but did not interrupt their subsequent sitting. The discrepancy

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

between this result and the one just mentioned [24] may be due to the many differences between these two studies including, the duration of the sitting period (4 hours *versus* 8.5 hours), the nature of the intervention (interrupted sitting only *versus* a interrupted sitting preceded by 30 minutes of exercise), or participant ages (mean 70 years *versus* 36 years).

Potential mechanisms involved in interrupted sitting on cognitive and cerebrovascular functions

Despite the inconsistent results reported, the available evidence suggested that there were acute beneficial effects of taking PA breaks on cognitive and cerebrovascular functions resulting from interrupting periods of prolonged sitting. There are several potential mechanisms that might explain why interrupting periods of sitting might be beneficial to brain function.

First, research both with humans and animal models suggests that activation of the locus coeruleus and the associated release of norepinephrine serves an important role in influencing attentional state, which may modulate cognitive performance [35,36]. NE release is at least partly modulated by the amount of PA. Studies found that prolonged uninterrupted sitting was associated with increased levels of fatigue [37] and decreased plasma level of NE [23]. On the other hand, previous studies have shown that moderate-intensity exercise can induce NE, indexed using salivary alpha-amylase, and enhance cognitive performances [23,38]. Two of the studies reviewed employed either relatively light-intensity of frequent PA breaks (every 30 min), or moderate-intensify but less frequent (every 60 min) PA breaks, and found no effects on either the plasma and urinary levels of NE or cognitive performance [23,28]. Moderate-intensity PA breaks every 30 minutes, progressively increasing duration of breaks, even at light-intensity, or adding an additional 30 minutes of continuous exercise before light-intensity PA breaks, all resulted in better cognitive performance [22,26]. It is known that cognitive performance is enhanced with increasing levels of NE [30], which, in turn, is affected by PA. Thus, individuals

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

interested in taking regular physical activity breaks from sitting for cognitive benefits should take into account the combination of intensity, frequency, and duration.

Secondly, glucose plays a vital role in supplying energy to cranial nerves and so glucose metabolism may influence brain function. The glucose-centric hypothesis postulates that exposure to acute *hyperglycemia* might decrease cognition [39]. Indeed, young and middle-aged people with diabetes have shown impaired performance on measures of memory, visual perception, and attention. This may be partly due to brain injury through neurodegeneration, frontal and hippocampal atrophy, and injury to white matter microstructure [40]. Studies have demonstrated that acute glucose ingestion (i.e., blood glucose concentrations increasing from 90 to 142 mg/dl) reduces the global cerebral metabolic rate of oxygen [41] and regional cerebral blood flow (CBF) [42], which is thought to result in poorer cognitive performance [43]. Regularly interrupting periods of sitting has been shown to reduce abruptly elevated postprandial glucose concentrations [44,45]. One of the studies included in this paper found that in healthy, normal-weight adults, taking short PA breaks frequently (every 30 minutes) resulted in higher CBF than less frequent (every 120 minutes) but longer PA breaks. This is in line with previous study examining the acute effects of taking regular activity breaks with different bouts of PA on blood glucose [46]. These suggested that regularly taking activity breaks after meals might be beneficial to the cognitive functions by preventing the reduction of cerebral oxygen, regional CBF, as well as repeated subtle brain injuries. On the other hand, it has been shown that *hypoglycemia* (from 100 to 48 mg/dl) also acutely reduces cortical blood oxygen level-dependent activation and cognitive function [47] as the brain requires an high level of energy supply [48]. Therefore, to avoid hyperinsulinemic hypoglycemia after meal consumption, taking sitting breaks might be a useful means of maintaining glucose concentrations within a stable range. In summary, studies suggest that taking breaks from sitting might benefit brain function by means of affecting glucose metabolism.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Limitations and the suggestions for the future study

Some caution is warranted when interpreting the findings of this review. First, the results cannot be generalised beyond the specific populations studied, i.e. children, adolescents, and individuals with chronic disease such as type 2 diabetes. The current literature has not focused on school-aged children. School-aged children spend the majority of class time sitting and this might negatively affect their cognitive function (i.e., attention state) [49]. In addition, individuals with type 2 diabetes showed impaired glucose metabolism that might, according to the glucose-centric hypothesis, modulate the relationship between PA and cerebrovascular function [39]. Thus, these populations need to be independently examined. Secondly, the majority of studies focussed on EFs (inhibitory control, working memory, or cognitive flexibility) but effects on other cognitive domains, such as problem solving and reasoning, as well as the precise cortical activation (i.e., neuro-electrophysiological responses) are still unknown. Thirdly, given the heterogeneity of exercise protocols, more studies are needed to clarify the effect of different types of interrupted sitting (frequency, intensity, duration and type) on brain function so as to provide more specific recommendations for public health. Finally, there is a need to further to explore the underlying mechanisms with the use of biochemical and physiological measures (such as neural activity in the locus coeruleus norepinephrine system and glucose metabolism etc.).

Conclusions

The current review provides preliminary evidence that interrupting prolonged periods of sitting might beneficially affect cognitive and cerebrovascular function in healthy adults. Given the diverse nature of study protocols used and the small number of available studies, there is a need for further work to examine which duration, frequency, and intensity of sitting breaks optimal for cognitive performance.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Funding information:

This study was financially supported by a joint grant from the "Institute for Research Excellence in Learning Sciences" at National Taiwan Normal University (NTNU) in the Featured Areas Research Center Program within the framework of the Higher Education Sprout Project by the Ministry of Education (MOE).

Conflict of interest: There is no conflict of interest to report.

Author Contributions

TYC conceived the study concept, contributed to the design of the study, literature search, data screening and extraction, assessed article quality, and wrote the first draft of the manuscript. YCC conceived the study concept, contributed to the design of the study, contributed to the literature search, data screening and extraction, assessed article quality, and critically reviewed and revised the manuscript. TMH conceived the study concept, contributed to the design of the study, provided assistance with data screening and extraction, and quality assessment, and critically reviewed and revised the manuscript. All authors approved the submission.

Acknowledgements

This study was financially supported by a joint grant from the "Institute for Research Excellence in Learning Sciences" at National Taiwan Normal University (NTNU) in the Featured Areas Research Center Program within the framework of the Higher Education Sprout Project by the Ministry of Education (MOE).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Reference

1 Sachdev, P. S. *et al.* Classifying neurocognitive disorders: the DSM-5 approach. *Nature Reviews Neurology* **10**, 634 (2014).

2 Willie, C. K., Tzeng, Y. C., Fisher, J. A. & Ainslie, P. N. Integrative regulation of human brain blood flow. *The Journal of physiology* **592**, 841-859 (2014).

3 Bertsch, K. *et al.* Resting cerebral blood flow, attention, and aging. *Brain research* **1267**, 77-88 (2009).

4 Wolters, F. J. *et al.* Cerebral perfusion and the risk of dementia: a population-based study. *Circulation* **136**, 719-728 (2017).

5 Keage, H. A. *et al.* Cerebrovascular function in aging and dementia: a systematic review of transcranial Doppler studies. *Dementia and geriatric cognitive disorders extra* **2**, 258-270 (2012).

6 Castro, O., Bennie, J., Vergeer, I., Bosselut, G. & Biddle, S. J. How sedentary are university students? A systematic review and meta-analysis. *Prevention Science* **21**, 332-343 (2020).

7 Van Dyck, D. *et al.* International study of objectively measured physical activity and sedentary time with body mass index and obesity: IPEN adult study. *International Journal of Obesity* **39**, 199-207 (2015).

8 Bakrania, K. *et al.* Associations between sedentary behaviors and cognitive function: Cross-sectional and prospective findings from the UK biobank. *American journal of epidemiology* **187**, 441-454 (2017).

9 Falck, R. S., Davis, J. C. & Liu-Ambrose, T. What is the association between sedentary behaviour and cognitive function? A systematic review. *Br J Sports Med* **51**, 800-811 (2017).

10 Vancampfort, D. *et al.* Mild cognitive impairment and sedentary behavior: a multinational study. *Experimental gerontology* **108**, 174-180 (2018).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

11 Triglav, J. *et al.* Physiological and cognitive measures during prolonged sitting: Comparisons between a standard and multi-axial office chair. *Applied ergonomics* **78**, 176-183 (2019).

12 Baker, R., Coenen, P., Howie, E., Williamson, A. & Straker, L. The short term musculoskeletal and cognitive effects of prolonged sitting during office computer work. *International journal of environmental research and public health* **15**, 1678 (2018).

13 Perdomo, S. J., Gibbs, B. B., Kowalsky, R. J., Taormina, J. M. & Balzer, J. R. Effects of alternating standing and sitting compared to prolonged sitting on cerebrovascular hemodynamics. *Sport Sciences for Health* **15**, 375-383 (2019).

14 Oberste, M. *et al.* Effects and moderators of acute aerobic exercise on subsequent interference control: a systematic review and meta-analysis Running title: Acute exercise and interference control. *Frontiers in Psychology* **10**, 2616 (2019).

15 Ludyga, S., Gerber, M., Brand, S., Holsboer-Trachsler, E. & Pühse, U. Acute effects of moderate aerobic exercise on specific aspects of executive function in different age and fitness groups: A meta-analysis. *Psychophysiology* **53**, 1611-1626 (2016).

16 Dempsey, P. C., Owen, N., Biddle, S. J. & Dunstan, D. W. Managing sedentary behavior to reduce the risk of diabetes and cardiovascular disease. *Current diabetes reports* **14**, 522 (2014).

17 Moher, D. *et al.* Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic reviews* **4**, 1 (2015).

18 Cohen, J. Statistical Power Analysis for the Behavioral Sciences–Second Edition. 12 Lawrence Erlbaum Associates Inc. *Hillsdale, New Jersey* **13** (1988).

19 Maher, C. G., Sherrington, C., Herbert, R. D., Moseley, A. M. & Elkins, M. Reliability of the PEDro scale for rating quality of randomized controlled trials. *Physical therapy* **83**, 713-721 (2003).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

20 Wheeler, M. J. *et al.* Morning exercise mitigates the impact of prolonged sitting on cerebral blood flow in older adults. *Journal of Applied Physiology* **126**, 1049-1055 (2019).

21 Wheeler, M. J. *et al.* Distinct effects of acute exercise and breaks in sitting on working memory and executive function in older adults: a three-arm, randomised cross-over trial to evaluate the effects of exercise with and without breaks in sitting on cognition. *Br J Sports Med* **54**, 776-781 (2020).

22 Mullane, S. L., Buman, M. P., Zeigler, Z. S., Crespo, N. C. & Gaesser, G. A. Acute effects on cognitive performance following bouts of standing and light-intensity physical activity in a simulated workplace environment. *J Sci Med Sport* **20**, 489-493 (2017).

23 Wennberg, P. *et al.* Acute effects of breaking up prolonged sitting on fatigue and cognition: a pilot study. *BMJ open* **6** (2016).

24 Carter, S. E. *et al.* Regular walking breaks prevent the decline in cerebral blood flow associated with prolonged sitting. *Journal of Applied Physiology* **125**, 790-798 (2018).

25 Vincent, G. E. *et al.* The impact of breaking up prolonged sitting on glucose metabolism and cognitive function when sleep is restricted. *Neurobiology of sleep and circadian rhythms* **4**, 17-23 (2018).

26 Chrismas, B. C., Taylor, L., Cherif, A., Sayegh, S. & Bailey, D. P. Breaking up prolonged sitting with moderate-intensity walking improves attention and executive function in Qatari females. *PloS one* **14**, e0219565 (2019).

27 Stoner, L. *et al.* Effects of acute prolonged sitting on cerebral perfusion and executive function in young adults: A randomized cross-over trial. *Psychophysiology* **56**, e13457 (2019).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

28 Bergouignan, A. *et al.* Effect of frequent interruptions of prolonged sitting on self-perceived levels of energy, mood, food cravings and cognitive function. *International Journal of Behavioral Nutrition and Physical Activity* **13**, 1-12 (2016).

29 Chang, Y., Labban, J., Gapin, J. & Etnier, J. The effects of acute exercise on cognitive performance: a meta-analysis. *Brain research* **1453**, 87-101 (2012).

30 Berridge, C. W. & Spencer, R. C. Differential cognitive actions of norepinephrine a2 and a1 receptor signaling in the prefrontal cortex. *Brain research* **1641**, 189-196 (2016).

31 American College of Sports Medicine. *ACSM's guidelines for exercise testing and prescription (10th ed.)*. (Lippincott williams & wilkins, 2018).

32 Ainslie, P. N. *et al.* Elevation in cerebral blood flow velocity with aerobic fitness throughout healthy human ageing. *The Journal of physiology* **586**, 4005-4010 (2008).

33 Sabayan, B. *et al.* Cerebrovascular hemodynamics in Alzheimer's disease and vascular dementia: a meta-analysis of transcranial Doppler studies. *Ageing research reviews* **11**, 271-277 (2012).

34 Conroy, D. A., Spielman, A. J. & Scott, R. Q. Daily rhythm of cerebral blood flow velocity. *Journal of circadian rhythms* **3**, 3 (2005).

35 Aston-Jones, G. & Cohen, J. D. An integrative theory of locus coeruleus-norepinephrine function: adaptive gain and optimal performance. *Annu. Rev. Neurosci.* **28**, 403-450 (2005).

36 Gelbard-Sagiv, H., Magidov, E., Sharon, H., Hendler, T. & Nir, Y. Noradrenaline modulates visual perception and late visually evoked activity. *Current Biology* **28**, 2239-2249. e2236 (2018).

37 Dempsey, P. C. *et al.* Prolonged uninterrupted sitting increases fatigue in type 2 diabetes. *Diabetes research and clinical practice* **135**, 128-133 (2018).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

38 Segal, S. K., Cotman, C. W. & Cahill, L. F. Exercise-induced noradrenergic activation enhances memory consolidation in both normal aging and patients with amnestic mild cognitive impairment. *Journal of Alzheimer's disease* **32**, 1011-1018 (2012).

39 Wheeler, M. J. *et al.* Sedentary behavior as a risk factor for cognitive decline? A focus on the influence of glycemic control in brain health. *Alzheimer's & Dementia: Translational Research & Clinical Interventions* **3**, 291-300 (2017).

40 Weinstein, G. *et al.* Glucose indices are associated with cognitive and structural brain measures in young adults. *Neurology* **84**, 2329-2337 (2015).

41 Xu, F. *et al.* Acute effect of glucose on cerebral blood flow, blood oxygenation, and oxidative metabolism. *Human brain mapping* **36**, 707-716 (2015).

42 Page, K. A. *et al.* Effects of fructose vs glucose on regional cerebral blood flow in brain regions involved with appetite and reward pathways. *Jama* **309**, 63-70 (2013).

43 Marshall, R. S. *et al.* Recovery of brain function during induced cerebral hypoperfusion. *Brain* **124**, 1208-1217 (2001).

44 Chen, Y.-C., Betts, J. A., Walhin, J.-P. & Thompson, D. Adipose Tissue Responses to Breaking Sitting in Men and Women with Central Adiposity. *Medicine and science in sports and exercise* **50**, 2049-2057 (2018).

45 Saunders, T. J. *et al.* The acute metabolic and vascular impact of interrupting prolonged sitting: a systematic review and meta-analysis. *Sports medicine* **48**, 2347-2366 (2018).

46 Peddie, M. C. *et al.* Breaking prolonged sitting reduces postprandial glycemia in healthy, normal-weight adults: a randomized crossover trial. *The American journal of clinical nutrition* **98**, 358-366 (2013).

47 Rosenthal, J. M. *et al.* The effect of acute hypoglycemia on brain function and activation: a functional magnetic resonance imaging study. *Diabetes* **50**, 1618-1626 (2001).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

48 Rolfe, D. & Brown, G. C. Cellular energy utilization and molecular origin of standard metabolic rate in mammals. *Physiological reviews* **77**, 731-758 (1997).

49 Pontifex, M. B., Parks, A. C., Henning, D. A. & Kamijo, K. Single bouts of exercise selectively sustain attentional processes. *Psychophysiology* **52**, 618-625 (2015).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and Cognition

Table 1 Characteristics of the studies included

Author/Country	Participants	Design	Arms	Outcomes	Results
Cognitive performances					
Stoner, et al. (2019)/ Hill	N = 20 (female = 14) Age: 18 to 35 y (M = 22 ± 3 y); Weight status: normal/overweight (BMI = 25.5 ± 6.1 kg/m^2); MVPA < 150 min per week; sitting time (M = 8.2 ± 2.2 h/day); sitting time: NR	Randomized cross-over	Trial started: 6–10 AM #**1. SIT:** sitting (3 h). #**2. BR:** sitting interrupted by 10 calf raises at the 15 min and 20 min points, and then every 10 min during the following sitting time; total PA = ~9 min.	EF-IC (stroop)	**EF-IC:** BR < SIT (ES = -0.85)
Wheeler, Green, et al. (2020) /Australia	N = 67 (female = 35) Age: ≥ 55 to ≤ 80 y (M = 67 ± 7 y); Weight status: overweight/obese (BMI: ≥ 25 to < 45 kg/m^2); MVPA < 150 min per week; sitting time > 5 h/day	Randomized cross-over	Trial started: ~8 AM #**1. SIT:** sitting (8 h). **2. EX+SIT:** sitting (1 h), continuous moderate-intensity (65–75% of age predicted HR max) walking (30 min), sitting (6.5 h); total PA = 30 min. #**3. EX+BR:** sitting (1 h), continuous moderate-intensity (65–75% of age predicted HR max) walking (30 min), sitting interrupted every 30 min with 3	1. Psychomotor function and speed of processing (detection test) 2. Attention (identification test) 3.Visual learning (one card learning test) 4. EF (groton maze learning test) 5.WM (one back test and two back test)	Psychomotor function & Attention & Visual learning & EF: EX + BR = SIT **WM:** EX+BR > SIT (ES = 0.25)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and Cognition

			min of light-intensity (3.2 km/h) walking (6.5 h); total PA = 69 min.		
Chrismas et al. (2019) /UK	N = 17 (female = 17) Age: 21–24 y (M = NR) Weight status & MVPA: NR; sitting time: > 7 h/day	Randomized cross-over	Trial started: ~8 AM #**1. SIT:** sitting (5 h). #**2. BR:** sitting interrupted every 30 min with 3 min of moderate-intensity walking (6.0 km/h, RPE = 12–14) walking; total PA = 27 min.	1. Attention & vigilance (rapid visual information processing and choice reaction time) 2. EF-IC (stroop) 3. WM (serial-3 and 7 subtractions)	**Attention & vigilance:** BR > SIT (ES = 0.89) **EF- IC:** BR > SIT (ES = 0.75) **EF-WM**: BR = SIT
Vincent et al (2018) /Australia	N = 6 (female = 0) Age: 20–35 y (M: 27 ± 4) Weight status: lean (waist circumference = ~ 84 ± 7 cm); MVPA & sitting time: NR	Randomized cross-over	#**1. SIT:** sitting (7 h); #**2. BR:** sitting interrupted every 30 min with 3 min of light-intensity walking (3.2 km/h); total PA = 51 min.	WM (digit symbol substitution test)	**No significant effects**
Mullane, Buman, Zeigler, Crespo and Gaesser (2017) /USA	N = 9 (female = 7) Age: 18–58 y (M = 30 ± 15 y) Weight status: overweight/obese (BMI = 29 ± 3 kg/m^2); MVPA < 150	Randomized cross-over	Trial started: 8 AM #**1. SIT:** sitting (8 h). **2. STAND:** standing #**3. CYCLE:** cycling (work rate = 20W, cadence (25–30 rpm). #**4. WALK:** walking (1.6 km/h)	1. Psychomotor function (the detection test) 2. WM (one back test) 3. EF (set-shifting Test)	**Psychomotor function:** CYCLE > SIT (ES = 1.1) WALK > SIT (ES = 0.42) CYCLE > WALK (ES = 0.63) **EF-WM:**

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	min per week, sitting time: NR		STAND, CYCLE & WALK for a predetermined time hourly: 10 min at 08:50 & 09:50, 15 min at 10:45 & 11:45, 20 min at 12:40 & 13:20, and 30 min at 14:00 & 15:30; total PA = 150 min.		CYCLE > SIT (ES = 0.91) WALK > SIT (ES = 0.74) **EF-CF:** CYCLE > SIT (ES = 0.78) WALK > SIT (ES = 0.66)
Bergouignan et al. (2016) /USA	N = 30 (female = 9) Age: 25–50 y (M = 31 ± 6 y) Weight status: non-obese (BMI = 18.5 to 29.9 kg/m^2); MVPA < 150 min/week; sitting time > 9 h/day	Randomized cross-over	Trial started: 8 AM #**1. SIT:** sitting (6 h). **2. EX:** continuous moderate-intensity (RPE = 12 to 13) walking (30 min); total PA = 30 min. #**3. BR:** sitting interrupted every 1 h with 5 min of moderate-intensity walking (RPE = 12 to 13); total PA = 30 min.	EF (flanker & trail-making test)	**No significant effects**
Wennberg et al. (2016) /Australia	N = 19 (female = 9) Age: 45–75 y (M = 60 ± 8 y) Weight status: overweight/obese (BMI: ≥ 25	Randomized cross-over	Trial started: 7–8 AM #**1. SIT:** sitting (7 h). #**2. BR:** sitting interrupted every 30 min with 3 min of light-intensity	1. EF (flanker & stroop) 2. Episodic memory (a face-name association test)	**No significant effects**

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and Cognition

	to ≤ 45 kg/m^2); MVPA < 150 min/week; sitting time > 5 h/day		walking (3.2 km/h, RPE = 9); total PA = 30 min.	3. WM (n-back & letter memory)	
Cerebrovascular responses					
Stoner, et al. (2019)/ Hill	N = 20 (female = 14) Age: 18 to 35 y (M = 22 ± 3 y); Weight status: normal/overweight (BMI = 25.5 ± 6.1 kg/m^2); MVPA < 150 min per week; sitting time (M = 8.2 ± 2.2 h/day); sitting time: NR	Randomized cross-over	Trial started: 6–10 AM #**1. SIT:** sitting (3 h). #**2. BR:** sitting interrupted by 10 calf raises at the 15 min and 20 min points, and then every 10 min during the following sitting time; total PA = ~9 min.	Prefrontal cortex perfusion	BR < SIT (ES = -0.97)
Wheeler, Dunstan, et al. (2019) /Australia	N = 12 (female = 2) Age: ≥ 55 to ≤ 80 y (M = 70 ± 7 y) Weight status: overweight/obese (BMI: ≥ 25 to < 45 kg/m^2); MVPA < 150 min/week; sitting time > 5 h/day	Randomized cross-over	Trial started: ~8 AM #**1. SIT:** sitting (8 h). **2. EX+SIT:** sitting (1 h), continuous moderate-intensity (65–75% of age predicted HR max) walking (30 min), prolonged sitting (6.5 h); total PA = 30 min. #**3. EX+BR:** sitting (1 h), continuous moderate-intensity (65–75% of age predicted HR max) walking (30 min), sitting interrupted every 30 min with 3	1. Temporal MCAv 2. Mean MCAv	**Temporal MCAv** SIT (negative linear trend, $p < 0.001$) EX+BR (positive linear trend, $p < 0.001$) **Mean MCAv** EX+BR = SIT

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and Cognition

			min of light-intensity (3.2 km/h) walking (6.5 h); total PA = 69 min.		
Carter et al. (2018) /UK	N = 15 (female = 5) Age: 36 ± 10 y Weight status: overweight (BMI = 25.5 ± 3.2 kg/m^2); MVPA = NR; sitting time (M = 8.2 ± 2.2 h/day)	Randomized cross-over	Trial started: 7–9 AM #**1. SIT:** sitting (4 h). #**2. Short-BR:** sitting interrupted every 30 min with 2 min of light-intensity (3.6 km/h) walking; total PA = 16 min. #**3. Long-BR:** sitting interrupted every 120 min with 8 min of light-intensity (3.6 km/h, RPE = ~9) walking; total PA = 16 min.	1. Supine-Mean MCAv 2. Seated-Mean MCAv	**Supine-Mean MCAv:** Short-BR > SIT (ES = 1.58) **Seated-Mean MCAv:** Short-BR > SIT (ES = 1.15) Short-BR > Long-BR (ES = 0.78)

Abbreviations: M = mean; MVPA = moderate-to-vigorous intensity physical activity; SIT = sitting; EX = continuous exercise; BR = breaking sitting; PA = physical activity; BMI = body mass index; CYCLE = cycling; NR = not report; Rating of perceived exertion (Borg scale) = RPE; IC = inhibitory control; WM = working memory; EF = executive function; MCAv = middle cerebral artery blood velocity; # = included arms for review; **In the result column:** > = positive effect from breaking sitting ($p < 0.05$); < = negative effect from breaking sitting ($p < 0.05$); ES = effect size (standardized mean difference)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 2 Quality assessment of included studies

Study	Eligibility criteria and source	Random allocation	Concealed allocation	Baseline comparability	Blinding subjects	Blinding therapists	Blinding assessors	Completeness of follow-up	Intention-to-treat	Between-group comparison	Point and variability measures	Total score
Wheeler et al. (2020)	Yes	Yes	Yes	No	No	No	No	Yes	No	Yes	Yes	5
Stoner et al. (2019)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Wheeler et al. (2019)	Yes	Yes	Yes	No	No	No	No	Yes	No	Yes	Yes	5
Sperlich et al. (2019)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Carter et al. (2018)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Mullane et al. (2017)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Bergouignan et al. (2016)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Wennberg et al. (2016)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Vincent et al. (2018)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Mean score (SD)	4.2 (0.4)											

PEDro scoring: Yes = 1, No = 0, NR = 0; Item 1 (eligibility criteria and source) is not used to calculate the PEDro score; the following cut-points were used to describe the quality of papers: 9–10 (excellent), 6–8 (good), 4–5 (fair), ≤ 3 (poor)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Reporting checklist for systematic review and meta-analysis.

Based on the PRISMA guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMAreporting guidelines, and cite them as:

Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement

		Reporting Item	Page Number
Title		The Acute Effect of Breaking up Prolonged Sitting on Cognitive and Cerebrovascular Function: A Systematic Review	pp.1
	#1	Identify the report as a systematic review, meta-analysis, or both.	
Abstract			pp.2
Structured summary	#2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	
Introduction			
Rationale	#3	Describe the rationale for the review in the context of what is already known.	p. 1-2

For peer review only

Objectives	#4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	p.1-2
Methods			
Protocol and registration	#5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address) and, if available, provide registration information including the registration number.	p. 2
Eligibility criteria	#6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rational	p. 2
Information sources	#7	Describe all information sources in the search (e.g., databases with dates of coverage, contact with study authors to identify additional studies) and date last searched.	p. 3
Search	#8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	p. 3
Study selection	#9	State the process for selecting studies (i.e., for screening, for determining eligibility, for inclusion in the systematic review, and, if applicable, for inclusion in the meta-analysis).	p. 3
Data collection process	#10	Describe the method of data extraction from reports (e.g., piloted forms, independently by two reviewers) and any processes for obtaining and confirming data from investigators.	p. 4
Data items	#11	List and define all variables for which data were sought (e.g., PICOS, funding sources), and any assumptions and simplifications made.	p. 2
Risk of bias in individual studies	#12	Describe methods used for assessing risk of bias in individual studies (including specification of whether this was done at the study or outcome level, or both), and how this information is to be used in any data synthesis.	p. 5
Summary measures	#13	State the principal summary measures (e.g., risk ratio, difference in means).	NR
Planned methods of analyis	#14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I2) for each meta-analysis.	

For peer review only

Risk of bias across studies	#15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NR
Additional analyses	#16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NR
Results			
Study selection	#17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	p. 6
Study characteristics	#18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citation.	p. 6
Risk of bias within studies	#19	Present data on risk of bias of each study and, if available, any outcome-level assessment (see Item 12).	p.8-9
Results of individual studies	#20	For all outcomes considered (benefits and harms), present, for each study: (a) simple summary data for each intervention group and (b) effect estimates and confidence intervals, ideally with a forest plot.	p. 7-9
Synthesis of results	#21	Present the main results of the review. If meta-analyses are done, include for each, confidence intervals and measures of consistency.	p. 7-9
Risk of bias across studies	#22	Present results of any assessment of risk of bias across studies (see Item 15).	NR
Additional analysis	#23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NR
Discussion			
Summary of Evidence	#24	Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., health care providers, users, and policy makers	p. 10
Limitations	#25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias).	p. 16
Conclusions	#26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	p. 16

For peer review only

Funding

Funding	#27	Describe sources of funding or other support (e.g., supply of data) for the systematic review; role of funders for the systematic review.	p. 17

None The PRISMA checklist is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist can be completed online using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

For peer review only

BMJ Open

The Acute Effect of Breaking up Prolonged Sitting on Cognition: A Systematic Review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-050458.R1
Article Type:	Original research
Date Submitted by the Author:	24-Nov-2021
Complete List of Authors:	Chueh, Ting-Yu; National Taiwan Normal University, Physical Education and Sport Sciences Chen, Yung-Chih; National Taiwan Normal University, Physical Education and Sport Sciences Hung , Tsung-Min ; National Taiwan Normal University, Physical Education and Sport Sciences
Primary Subject Heading:	Mental health
Secondary Subject Heading:	Mental health, Public health, Sports and exercise medicine
Keywords:	MENTAL HEALTH, PUBLIC HEALTH, SPORTS MEDICINE

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Breaking sitting and cognition

Title: The Acute Effect of Breaking up Prolonged Sitting on Cognition: A Systematic Review

Running head: Breaks from sitting and cognition

Word count: 4681

Authors:
Ting-Yu Chueh[1], Yung-Chih Chen*[1], Tsung-Min Hung*[1, 2]

Affiliations:
[1] Department of Physical Education and Sport Sciences, National Taiwan Normal University, Taipei, Taiwan
[2] Institute for Research Excellence and Learning Sciences, National Taiwan Normal University, Taipei, Taiwan

***Correspondence:**
Yung-Chih Chen, Ph.D.
E-mail: yc.chen@ntnu.edu.tw
Department of Physical Education and Sport Sciences, National Taiwan Normal University, Taipei, Taiwan
162, Section 1, Heping E. Rd., Taipei City 106, Taiwan

Tsung-Min Hung, Ph.D., FNAK, FISSP
E-mail: ernesthungkimo@yahoo.com.tw
Department of Physical Education and Sport Sciences & Institute for Research Excellence and Learning Sciences, National Taiwan Normal University, Taipei, Taiwan
162, Section 1, Heping E. Rd., Taipei City 106, Taiwan

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Abstract

Objectives: To review the current evidence on the acute effects of interrupting prolonged periods of sitting with intermittent physical activity (PA) on cognition in healthy populations.

Design: This systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.

Methods: Studies were included if they investigated the acute effects of taking regular PA breaks from sitting on cognition in healthy populations without any cardiovascular disease, history of brain injury, or psychiatric or neurological disorder. Four electronic databases—PubMed, Scopus, MEDLINE, and ProQuest—were searched for eligible studies on September 20, 2020. Study quality was assessed using the Physiotherapy Evidence Database scale. This review is registered in PROSPERO (number: CRD42020147536).

Results: Seven studies, involving 168 participants aged between 18 and 80 years, were eligible for inclusion in this review. Three of the seven studies found positive effects of interrupting sitting with either a) 3 min of relatively high-intensity (6 km/h) walking every 30 min on attention and inhibitory control in young adults; b) hourly breaks with progressively longer durations (10–30 min) of very light-intensity cycling/walking on attention, working memory, and cognitive flexibility in adults with obesity; or c) an initial bout of continuous moderate-intensity exercise, followed by interruption of post-exercise sitting with 3-min breaks of light-intensity walking (3.2 km/h) every 30 min, on working memory in older adults with overweight.

Conclusion: The small number of available studies provided preliminary results that interrupting periods of sitting may beneficially affect cognition in healthy adults. Given the limited evidence on this topic in the literature and the heterogeneity of PA protocols across the included studies, our results regarding the effectiveness of interrupting prolonged sitting with PA breaks in improving cognition warrant further verification.

Funding: Higher Education Sprout Project by Ministry of Education. Award/Grant number is not applicable.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Key words: Sedentary behavior; Cognition; Physical activity; Breaking sitting; Brain health

For peer review only

Strengths and limitations of this study

- ✓ This is the first study to systematically review the acute effects of taking regular physical activity breaks from sitting on cognition in healthy individuals.
- ✓ This review highlights significant research gaps in the field.
- ✓ This review identifies potential mechanisms underlying the acute effects of taking physical activity breaks from sitting on cognition.
- ✓ A meta-analysis is not possible due to the methodological heterogeneity of the included studies.
- ✓ Firm conclusions are limited by the small number of available studies.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Introduction

Optimal cognitive performance in all cognitive domains, including attention, executive functions (EFs), and memory engagement, is important for optimizing daily functioning and reducing cognitive decline, which are critical to productivity and quality of life [1,2]. In classrooms and offices, for example, individuals require attention to focus on relevant information while relying on distinct EFs to suppress irrelevant distraction (inhibitory control), hold and mentally organize information (working memory [WM]), and adjust their behavior or thoughts based on updated demands, rules, or priorities (cognitive flexibility) to achieve goals [3-6]. These abilities have profound attribution effects on academic attainment and job performance [7,8]. In addition, a healthy cognitive state is beneficial in late life, because it reduces age-related cognitive decline [9]. Therefore, it is important to maintain and/or enhance cognition in daily life.

Sedentary behavior and physical activity (PA) are two important modifiable factors associated with cognition [10,11]. Adults typically spend the vast majority (8–10 h on average) of their waking day in sedentary behavior such as sitting and/or reclining [12,13]. Recent studies have indicated that increased sedentary behavior was associated with lower cognitive performance [14-16]. This association was partially independent of the PA level[17]. Studies also have demonstrated that prolonged sitting transiently deteriorated cognitive performance (i.e., increases problem solving and attention task errors) [18,19]. Although it has been well established that cognitive performance is enhanced after 20 to 30 min of continuous/structured exercise [20,21], individuals who engage in such exercise may be largely sedentary throughout the rest of the day [22]. Accordingly, taking regular PA breaks from sitting throughout the day may be a feasible strategy to offset the negative effects of prolonged sitting on cognition.

Taking regular PA breaks from sitting has been shown to be a practical approach to reduce the negative effects of acute prolonged sitting on cardiovascular and metabolic functions [23-25].

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Given that acute uninterrupted sitting has negative effects on cognition [18,19], related studies have extended the investigation from the physiological outcomes to the cognitive outcomes of taking regular breaks from sitting. Although recent studies have investigated the effects of taking breaks from sitting on cognition, they have not yet been subjected to a systematic review, which can provide a high level of confidence in the current findings. Furthermore, understanding the physiological mechanisms behind the acute effects of taking PA breaks from sitting on cognition is important to facilitate the design of successful PA approaches. Thus, the aims of this paper were (1) to systematically review studies investigating the acute effects of interrupting prolonged periods of sitting on cognition compared with uninterrupted sitting; (2) to provide a methodological assessment of all of the reviewed studies; and (3) to discuss the potential underlying physiological mechanisms of any effects found and provide suggestions for future studies.

Methods

This review was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines [26,27] and is registered in the International Prospective Register of Systematic Reviews (PROSPERO; identification code: CRD42020147536) [1].

Eligibility criteria

The Population, Intervention, Comparison, Outcome, and Study (PICOS) framework was used to determine the inclusion criteria for studies [26], as follows. (P) Participants: participants

[1] This study was initially planned to include studies that examined the acute effects of interrupting sitting on cerebrovascular function. However, only three studies met the eligibility criteria in the final search, and two of them involved the same dataset of cognition studies. Thus, we changed the focus of this systematic review to studies that evaluated the acute effects of interrupting sitting on cognition.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

were healthy and without any cardiovascular disease, history of brain injury, or psychiatric or neurological disorder. (I) Intervention: interventions were of an acute nature, involving interrupting prolonged periods of sitting with a regular pattern of PA at predefined intervals (e.g., every 30 min). (C) Comparator: studies included a non-treatment control group (i.e., the prolonged sitting condition). (O) Outcome: studies tested cognitive performance. (S) Study design: studies were randomized controlled or non-randomized controlled trials. Studies were excluded if they (1) involved only a single bout of PA in the intervention condition; (2) did not explicitly state the PA-related parameters, such as type, intensity, frequency, and duration; and/or (3) were not published in an English language peer-reviewed journal.

Information sources and search strategy

Four electronic databases—PubMed, Scopus, MEDLINE, and ProQuest—were searched for eligible studies from inception. The initial search was conducted on August 5, 2019, and the final search was completed on September 20, 2020. The key words used to search the titles and abstracts were discussed by the research team to maximize the chance of identifying relevant articles. The following key words were used: (*"Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting"*) *AND* (*cognitive OR cognition OR executive OR brain OR cerebral*). The full search terms used in all four databases are provided in Supplementary Appendix S1.

The inclusion and exclusion of articles were decided according to the PICOS criteria, with the screening and selection of studies being completed by two authors ([redacted]). First, the titles and abstracts were independently assessed by these two authors and initially coded as "yes," "no," or "maybe" for inclusion. The same two authors then reviewed the full texts of the "yes" and "maybe" studies, and disagreements regarding the inclusion of any study were resolved by discussion with a third reviewer ([redacted]). The reference lists of all of the included

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

articles were then searched to check for potentially relevant studies. **Figure 1** provides an overview of the selection process.

Data collection and items

The data collection was conducted by the same two authors ([redacted]). The authors thoroughly read the included studies and extracted the following data: (1) first author's name, publication year, and country of data collection; (2) participants' characteristics (viz., sample size, age, weight status, duration of exercise, and daily sedentary behavior); (3) study design; (4) details of the intervention protocols; and (5) outcome measures.

Synthesis methods

The results in this review were analyzed through a process of narrative synthesis. The data were first analyzed by one reviewer ([redacted]) and then verified by a second reviewer ([redacted]). As all of the necessary information could be obtained from the articles, no authors were contacted for information. A p-value of < 0.05, presented in the original studies, was used across the studies to determine the significant effects of interrupting sitting with PA on cognition. The results were converted to effect sizes (standardized mean difference) and entered into the results column of **Table 1**. Cohen's d was used to interpret the effect sizes: small ($d = 0.2$), medium ($d = 0.5$), and large ($d = 0.8$) [28].

Quality assessment

The quality of the included studies was assessed using the Physiotherapy Evidence Database (PEDro) scale [29], which produces a quantitative assessment of bias as a score out of 10, with higher scores indicating higher methodological quality. The PEDro scale assesses allocation bias (randomization, concealment of allocation), performance bias (blinding of participants and personnel), and detection bias (blinding of outcome assessment), in addition

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

to other items, including eligibility criteria, baseline comparability, retention rate, intention-to-treat, between-group statistical comparisons for at least one key outcome, point measures, and variability measures. When available, the PEDro score was obtained from the PEDro database. This was possible only for one study [30]. For studies for which the PEDro score was not available from the database, it was calculated by the two review authors ([redacted]) independently. Differences were resolved by discussion and adjudicated by a third author ([redacted]) as required.

Patient and public involvement

No patient involved.

Results

Study selection

In total, seven studies met all of the inclusion criteria and were included in this review. **Figure 1** illustrates the study selection process in a PRISMA flow diagram [26].

Study characteristics

All of the seven included studies were randomized crossover trials and comprised a total of 91 male and 77 female participants aged between 18 and 80 years. The body mass index (BMI) of the participants across all of the studies was between 18.5 and 44 kg/m^2. Most of the participants (57%) were overweight or obese (BMI > 25 kg/m^2) [30-32], and most (86%) did not report being actively engaged in regular exercise, defined as engaging in moderate to vigorous physical activity (MVPA) for at least 150 min per week. The duration of sedentary behavior across all studies was between 5 and 9 h per day. **Table 1** provides a summary of the study designs and participant characteristics.

Regarding the PA intensity during breaks from sitting, four out of the seven studies stated that the exercise was of light-intensity (walking on a treadmill at 1.6–3.6 km/h or cycling at 20

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

watts with 25–30 rpm) [30-33], two studies reported that the break consisted of moderate-intensity PA (walking on a treadmill at ~ 6 km/h) [34] and one study implemented 10 calf raises at a rate of 0.33 Hz (20 per minute) [35].

With regard to the duration and frequency of breaks, three studies implemented 3-min PA breaks from sitting every 30 min, for a total daily exercise duration of 27 [34], 30 [32], and 51 min [33]. In one study, every 60 min of sitting was interrupted with 5 min of PA, for a total daily PA duration of 30 min [36], and in another study, every 60 min of sitting was interrupted with progressively longer durations of either walking or cycling (i.e., 10, 15, 20, and 30 min), for a daily total of 150 min of very low-intensity PA [31]. One study instructed the participants to performed 10 calf raises within 30 s after every 5 or 10 min of sitting, for a total daily PA duration of 9 min (17 sets of calf exercise) [35]. The remaining study included an initial 30-min bout of moderate-intensity walking, followed by sitting interrupted every 30 min with 3 min of PA, for a total daily PA duration of 69 min [30]. **Table 1** summarizes the intervention characteristics for all of the included studies.

Results of individual studies

All of the included studies focused on EFs, three studies evaluated attention or processing speed [30,31,34], two assessed memory [31], and one study used a visual learning task [30]. Three studies found positive effects of PA breaks from sitting on EFs. In one study, compared with prolonged sitting, progressively longer durations of both light-intensity walking and cycling (i.e., 10, 15, 20, and 30 min) after every 60 min of sitting resulted in greater attention, WM, and cognitive flexibility, with small to large effect sizes (d = 0.62–0.91) [31]. Interestingly, cycling resulted in a better improvement in attention than walking, with a medium effect size (d = 0.63). Taking 3-min breaks of moderate-intensity PA (6.0 km/h, rating of perceived exertion [RPE] = 12–14) every 30 min resulted in improved inhibitory control with a large effect size (d = 0.74) [34], whereas taking 3-min breaks of light-intensity walking (3.2 km/h)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

every 30 min [32,33] or 5-min breaks of moderate-intensity walking (RPE = 12–13) every 60 min showed no significant effect on any cognitive performance measure [36]. Three-minute breaks of light-intensity PA (3.2 km/h) every 30 min after an initial bout of continuous moderate-intensity exercise (65–75% of the age-predicted maximal heart rate) had a beneficial effect on WM (d = 0.25) [30]. One study found adverse effects of intermittent 10 calf raise exercises on inhibitory control (d = -0.85) [35]. Two studies reported the positive effects of taking 3-min breaks of moderate-intensity PA every 30 min or progressively longer breaks of light-intensity walking and cycling every 60 min on attention, with medium to large effect sizes (d = 0.42 to 1.1) [31,34]. Interestingly, cycling resulted in a better improvement in attention than walking, with a medium effect size (d = 0.63). No significant effects of interrupting sitting with PA was found in other cognitive domains.

Methodological quality

The included studies had a mean PEDro score of 4.1, indicating "fair" methodological quality. All studies reported to have performed random allocation of participants to the trials, but only one study reported performing concealed allocation [30]. This suggests the presence of allocation biases in the other studies. None of the studies mentioned the blinding of participants and/or researchers because it is impossible to blind an intervention that requires the participants to perform activities. All of the studies provided a key outcome measure for more than 85% of their participants, and all of the studies adopted a crossover trial design, which meant that all participants received all of the designed treatments. Nevertheless, none of the studies explicitly stated whether at least one main outcome was included in the statistical analysis (intention-to-treat analysis) of all of the participants, regardless of any subsequent withdrawal from treatment. This suggests the presence of biased comparisons between the treatment arms. Finally, all of the studies provided statistical comparisons and valid measures for at least one key outcome measure of interest. Detailed results of the PEDro assessment are presented in **Table 2**.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Discussion

The purpose of this study was to review the results of studies that investigated whether interrupting prolonged periods of sitting with PA breaks would improve cognition. Seven studies were identified that investigated the acute effects of such interruptions in healthy individuals. All of these studies were published recently (between 2016 and 2020). Three of the seven studies provided preliminary evidence that taking regular PA breaks from sitting offers short-term benefits for cognition in healthy adults. However, no definite conclusions could be drawn regarding the effectiveness of taking PA breaks from sitting in improving cognition due to the heterogeneity of the PA protocols across the included studies and the small number of available studies.

The acute effects of taking PA breaks from sitting on cognition

This review suggests that taking PA breaks from sitting, specifically their intensity and/or volume, might play a potential role in regulation of cognitive performance. For example, compared with prolonged sitting, taking 3-min breaks of moderate-intensity walking (6 km/h) every 30 min was reported to result in better attention and inhibitory control in young female adults [34]. However, the same protocol (3 min of PA every 30 min) but with light-intensity walking (3.2 km/h) produced no effect on episodic memory, inhibitory control, or WM in overweight/obese middle-aged adults [32] or on attention or WM in young adults [33]. Studies have reported that exercise intensity may be an important determinant of cognitive benefits [37,38]. A meta-analysis demonstrated that the positive acute effect of exercise on cognition is seen when the exercise intensity is moderate, but the effect diminishes when the exercise intensity is very light [39]. One possible explanation is that greater exercise intensity is needed to induce sufficient

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

levels of norepinephrine (NE) to facilitate cognitive performance [40]. Thus, the intensity of PA during breaks from sitting may be one of the factors determining the effectiveness of the breaks.

However, taking higher-intensity PA breaks is not always beneficial if the frequency of such breaks is low. For example, another study of non-obese adults investigated the effects of a similar walking speed (~5.7 km/h), but less frequent (hourly) and slightly longer (5 min), walking breaks from sitting and found no effect on the measures of inhibitory control or cognitive flexibility [36]. This implies that the benefits of taking relatively high intensity PA breaks from sitting might be reduced by having infrequent breaks. Interestingly, taking hourly low-intensity PA breaks might still benefit cognition if these breaks are sufficiently long (i.e., 10, 15, 20, and 30 min). As indicated in one study [31], both very low-intensity cycling (work rate = 20 W, cadence = 25–30 rpm) and walking (1.6 km/h) resulted in better attention (the detection test), cognitive flexibility (the set-shifting test), and WM (the one-back test) compared with prolonged sitting. Despite the low intensity of PA breaks, the longer duration of PA breaks adopted in this study, which resulted in a greater amount of exercise (150 min in total), might have had a compensating effect. This finding implies that volume of exercise should be considered and carries important implication for some individuals who are unable to engage in moderate to vigorous intensity exercise. Collectively, these results suggest that the effects of taking breaks from sitting on cognitive function are dependent on the intensity, frequency, and duration of the PA performed during the breaks. Future studies are recommended to determine the minimum number and duration of low-intensity PA breaks from sitting required for a meaningful improvement in cognition.

The study that evaluated the effects of a morning bout of 30 min of moderate-intensity exercise with or without subsequent light-intensity PA breaks every 30 min on cognitive performance in obese/overweight older adults provides another fruitful line of research [30]. A single bout of moderate-intensity PA before engaging in prolonged sitting showed better EFs, but not WM, relative to the prolonged sitting condition, whereas the interruption of post-

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

exercise sitting with several additional bouts of light-intensity PA led to improved WM, but not EFs [30]. These results suggest that a single bout of continuous moderate-intensity exercise with or without subsequent PA breaks provide benefits in different cognitive domains. These divergent effects may be attributed to individual differences in the level of aerobic fitness and the timing and order of the administration of cognitive tests following the cessation of exercise. It is reasonable to assume that the participants in that study had lower aerobic fitness, based on the provided demographic data (old age, sedentary lifestyle, and obese/overweight status). In this study, the test of EFs was administered almost immediately (approximately 2 min) after the 3-min light-intensity PA breaks, followed by the WM test and other cognitive assessments. A previous meta-analytic review indicated that the largest effect of lower-intensity PA on cognitive performance was immediately after the PA, whereas that of higher-intensity activities was after a delay of ~10 min [39]. However, an individual's level of aerobic fitness may moderate the physiological demands and/or perception of the physical activity stimulus [41]. Chang, et al. [39] observed that aerobic fitness was a moderating variable when cognition was assessed immediately after the cessation of a bout of PA. Thus, the beneficial effects of a single bout of continuous moderate-intensity exercise with subsequent light-intensity PA breaks on EFs in this study may have been diminished due to the lower aerobic fitness level of the participants and the small time interval between exercise cessation and EF assessment. Moreover, when the aim is to maintain WM, it appears that it is necessary to avoid prolonged sitting even after an exercise session.

Of note, one study found adverse effects of frequent calf raise exercises on inhibitory control in young adults [35]. The impaired inhibitory control performance was accompanied by a decreased prefrontal cortex perfusion, as measured by near-infrared spectroscopy, supporting that prefrontal cortex activity could be a potential mechanism for modulating cognitive performance [6]. The negative effect of acute exercise on EFs was observed only after a bout of high-intensity exercise [39]. We found no study reporting a negative effect on cognition following

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

a short bout of low-intensity exercise, such as calf raises. The reason for this result is unknown; therefore, future studies should attempt to replicate this result and investigate the reason.

Taken together, the literature reviewed suggests that to understand the effects of PA breaks from prolonged sitting on cognitive performance, the effects of the frequency, intensity, and duration of the PA undertaken during the breaks need to be considered concurrently. The current evidence suggests that a) short, frequent breaks (at least every 30 min) involving moderate-intensity PA (walking at least 6 km/h); b) longer, hourly breaks involving even very low-intensity PA; and c) a single bout of 30-min continuous moderate-intensity exercise with subsequent short PA breaks are all beneficial to cognitive performance (e.g., attention, EFs, or memory-related tasks). However, the heterogeneity of PA protocols across the reviewed studies and the small number of available studies make direct comparisons of study results difficult. Furthermore, the findings of the effectiveness of PA breaks from sitting in improving cognition need to be further confirmed, because fewer than half of the studies included reported significant positive results.

Potential mechanisms underlying the effects of interrupting sitting on cognition

Despite the inconsistent results across the reviewed studies, the available evidence suggests that taking PA breaks from prolonged sitting exerts acute beneficial effects on cognition. Several potential mechanisms may explain why interrupting periods of sitting is beneficial to brain function.

First, research in both humans and animal models suggests that the activation of the locus coeruleus and the associated release of NE play important roles in influencing the attention state, which may modulate cognitive performance [42,43]. NE release is at least partly modulated by the amount of PA. Studies have found that prolonged uninterrupted sitting is associated with increased levels of fatigue [44] and decreased plasma levels of NE [32], whereas moderate-intensity exercise can induce NE release (indexed by salivary alpha-amylase concentrations) and

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

enhance cognitive performance [32,45]. Two studies that used either frequent breaks of relatively light-intensity PA (every 30 min) or less frequent breaks of moderate-intensity PA (every 60 min) found no beneficial effects on the plasma and urinary levels of NE or cognitive performance [32,36]. Moderate-intensity PA breaks every 30 min; progressively increasing the duration of breaks, even at light-intensity PA; and performing an initial bout of continuous moderate-intensity exercise for 30 min before subsequent light-intensity PA breaks during sitting, all resulted in improved cognitive performance compared with the prolonged sitting condition [31,34]. It is known that cognitive performance is enhanced with increasing levels of NE [40], which, in turn, is affected by PA. Thus, individuals interested in taking regular physical activity breaks from sitting for cognitive benefits should take into account the combination of intensity, frequency, and duration.

Secondly, glucose plays a vital role in supplying energy to cranial nerves and so glucose metabolism may influence cognition. The glucose-centric hypothesis postulates that exposure to acute hyperglycemia impairs cognition [11]. Indeed, young and middle-aged people with diabetes have shown impaired performance on measures of memory, visual perception, and attention. This may be partly due to brain injury through neurodegeneration, frontal and hippocampal atrophy, and injury to the white matter microstructure [46]. Studies have demonstrated that acute glucose ingestion (i.e., blood glucose concentrations increasing from 90 to 142 mg/dl) reduces the global cerebral metabolic rate of oxygen [47] and regional cerebral blood flow (CBF) [48], which result in poorer cognitive performance [49]. Regularly interrupting periods of sitting has been shown to reduce abrupt elevations in postprandial glucose concentrations [23,25]. One study found that in healthy, normal-weight adults, taking short but frequent PA breaks (every 30 min) resulted in higher CBF than taking longer but less frequent PA breaks (every 120 min). This is in line with the results of a previous study that examined the acute effects of taking regular PA breaks with different bouts of PA on blood glucose [50]. These findings suggest that regularly taking PA breaks after meals might be beneficial to

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

cognition by preventing both the reductions in cerebral oxygen and regional CBF and repeated subtle brain injuries. On the other hand, it has been shown that hypoglycemia (from 100 to 48 mg/dl) also acutely reduces blood oxygen level-dependent cortical activation and cognitive function [51], as the brain requires a high level of energy supply [52]. Therefore, to avoid hyperinsulinemic hypoglycemia after meal consumption, taking breaks from sitting may be a useful means of maintaining blood glucose concentration within a stable range. In summary, studies suggest that taking breaks from sitting might benefit cognition by means of affecting glucose metabolism.

Limitations and suggestions for future research

Some caution is warranted when interpreting the findings of this review. First, the results cannot be generalised beyond the specific populations studied, i.e. children, adolescents, and individuals with chronic disease such as type 2 diabetes. The current literature has not focused on school-aged children. School-aged children spend the majority of class time sitting and this might negatively affect their cognitive function (i.e., attention state) [53]. In addition, individuals with type 2 diabetes showed impaired glucose metabolism that might, according to the glucose-centric hypothesis, modulate the relationship between PA and cerebrovascular function [11]. Thus, these populations need to be independently examined. Second, given the heterogeneity of PA protocols, cognitive assessments, and sample size across the included studies, more studies with larger samples are needed to clarify the role of different PA characteristics (i.e., frequency, intensity, duration, and type) in the effect of PA breaks from sitting on distinct cognitive domains so as to provide more precise recommendations for public health. Third, this review exclusively focused on studies that assessed the acute effects of PA breaks on cognition. Future research should explore the accumulative effects of multiple days of sitting with or without PA breaks on cognition. Fourth, there is a need to further explore the underlying mechanisms using biochemical and physiological measures, such as neural activity in the locus coeruleus NE

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

system and glucose metabolism. Finally, future studies should examine the isolated contributions of interrupting prolonged sitting versus continuous PA to improvements in cognition.

Conclusions

The current review provides preliminary evidence that the intensity and/or volume of PA breaks from prolonged sitting may determine the effects of such breaks on cognition. However, due to the heterogeneity of PA protocols across studies and limited evidence, firm conclusions regarding the effectiveness of interrupting sitting in improving cognition cannot be drawn. Further studies are warranted to examine the duration, frequency, and intensity of PA breaks required to optimize cognitive performance.

Conflict of interest: There is no conflict of interest to report.

Ethics approval: Ethical approval was not applicable because this study retrieved data from already published studies.

Author contributions: [redacted] conceived the study concept; contributed to the study design, literature search, and data screening and extraction; assessed the article quality; and wrote the first draft of the manuscript. [redacted] conceived the study concept; contributed to the study design, literature search, and data screening and extraction; assessed the article quality; and critically reviewed and revised the manuscript. [redacted] conceived the study concept, contributed to the study design, assisted with data screening and extraction and quality assessment, and critically reviewed and revised the manuscript. All of the authors approved the submission.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

357

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

References

1 Hill, N. L. *et al.* Subjective cognitive impairment and quality of life: a systematic review. *International Psychogeriatrics* **29**, 1965-1977 (2017).

2 Sanz, J. H. *et al.* Executive function and psychosocial quality of life in school age children with congenital heart disease. *J Pediatr* **202**, 63-69 (2018).

3 Diamond, A. Executive functions. *Annu Rev Psychol* **64**, 135-168 (2013).

4 Miyake, A. *et al.* The unity and diversity of executive functions and their contributions to complex "frontal lobe" tasks: A latent variable analysis. *Cognitive psychology* **41**, 49-100 (2000).

5 Zelazo, P. D. Executive function: Reflection, iterative reprocessing, complexity, and the developing brain. *Developmental Review* **38**, 55-68 (2015).

6 Miller, E. K. & Cohen, J. D. An integrative theory of prefrontal cortex function. *Annual review of neuroscience* **24**, 167-202 (2001).

7 Lang, J. W., Kersting, M., Hülsheger, U. R. & Lang, J. General mental ability, narrower cognitive abilities, and job performance: The perspective of the nested-factors model of cognitive abilities. *Personnel psychology* **63**, 595-640 (2010).

8 Rohde, T. E. & Thompson, L. A. Predicting academic achievement with cognitive ability. *Intelligence* **35**, 83-92 (2007).

9 Buckner, R. L. Memory and executive function in aging and AD: multiple factors that cause decline and reserve factors that compensate. *Neuron* **44**, 195-208 (2004).

10 Erickson, K. I. *et al.* Physical Activity, Cognition, and Brain Outcomes: A Review of the 2018 Physical Activity Guidelines. *Medicine and science in sports and exercise* **51**, 1242-1251 (2019).

11 Wheeler, M. J. *et al.* Sedentary behavior as a risk factor for cognitive decline? A focus on the influence of glycemic control in brain health. *Alzheimer's & Dementia: Translational Research & Clinical Interventions* **3**, 291-300 (2017).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12 Castro, O., Bennie, J., Vergeer, I., Bosselut, G. & Biddle, S. J. How sedentary are university students? A systematic review and meta-analysis. *Prevention Science* **21**, 332-343 (2020).

13 Van Dyck, D. *et al.* International study of objectively measured physical activity and sedentary time with body mass index and obesity: IPEN adult study. *International Journal of Obesity* **39**, 199-207 (2015).

14 Bakrania, K. *et al.* Associations between sedentary behaviors and cognitive function: Cross-sectional and prospective findings from the UK biobank. *American journal of epidemiology* **187**, 441-454 (2017).

15 Falck, R. S., Davis, J. C. & Liu-Ambrose, T. What is the association between sedentary behaviour and cognitive function? A systematic review. *Br J Sports Med* **51**, 800-811 (2017).

16 Vancampfort, D. *et al.* Mild cognitive impairment and sedentary behavior: a multinational study. *Experimental gerontology* **108**, 174-180 (2018).

17 Ku, P.-W., Liu, Y.-T., Lo, M.-K., Chen, L.-J. & Stubbs, B. Higher levels of objectively measured sedentary behavior is associated with worse cognitive ability: Two-year follow-up study in community-dwelling older adults. *Experimental gerontology* **99**, 110-114 (2017).

18 Triglav, J. *et al.* Physiological and cognitive measures during prolonged sitting: Comparisons between a standard and multi-axial office chair. *Applied ergonomics* **78**, 176-183 (2019).

19 Baker, R., Coenen, P., Howie, E., Williamson, A. & Straker, L. The short term musculoskeletal and cognitive effects of prolonged sitting during office computer work. *International journal of environmental research and public health* **15**, 1678 (2018).

For peer review only

20 Oberste, M. *et al.* Effects and moderators of acute aerobic exercise on subsequent interference control: a systematic review and meta-analysis Running title: Acute exercise and interference control. *Frontiers in Psychology* **10**, 2616 (2019).

21 Ludyga, S., Gerber, M., Brand, S., Holsboer-Trachsler, E. & Pühse, U. Acute effects of moderate aerobic exercise on specific aspects of executive function in different age and fitness groups: A meta-analysis. *Psychophysiology* **53**, 1611-1626 (2016).

22 Dempsey, P. C., Owen, N., Biddle, S. J. & Dunstan, D. W. Managing sedentary behavior to reduce the risk of diabetes and cardiovascular disease. *Current diabetes reports* **14**, 522 (2014).

23 Chen, Y.-C., Betts, J. A., Walhin, J.-P. & Thompson, D. Adipose Tissue Responses to Breaking Sitting in Men and Women with Central Adiposity. *Medicine and science in sports and exercise* **50**, 2049-2057 (2018).

24 Dunstan, D. W. *et al.* Breaking up prolonged sitting reduces postprandial glucose and insulin responses. *Diabetes care* **35**, 976-983 (2012).

25 Saunders, T. J. *et al.* The acute metabolic and vascular impact of interrupting prolonged sitting: a systematic review and meta-analysis. *Sports medicine* **48**, 2347-2366 (2018).

26 Moher, D. *et al.* Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic reviews* **4**, 1 (2015).

27 Moher, D., Liberati, A., Tetzlaff, J. & Altman, D. G. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Int J Surg* **8**, 336-341 (2010).

28 Cohen, J. Statistical Power Analysis for the Behavioral Sciences–Second Edition. 12 Lawrence Erlbaum Associates Inc. *Hillsdale, New Jersey* **13** (1988).

29 Maher, C. G., Sherrington, C., Herbert, R. D., Moseley, A. M. & Elkins, M. Reliability of the PEDro scale for rating quality of randomized controlled trials. *Physical therapy* **83**, 713-721 (2003).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

30 Wheeler, M. J. *et al.* Distinct effects of acute exercise and breaks in sitting on working memory and executive function in older adults: a three-arm, randomised cross-over trial to evaluate the effects of exercise with and without breaks in sitting on cognition. *Br J Sports Med* **54**, 776-781 (2020).

31 Mullane, S. L., Buman, M. P., Zeigler, Z. S., Crespo, N. C. & Gaesser, G. A. Acute effects on cognitive performance following bouts of standing and light-intensity physical activity in a simulated workplace environment. *J Sci Med Sport* **20**, 489-493 (2017).

32 Wennberg, P. *et al.* Acute effects of breaking up prolonged sitting on fatigue and cognition: a pilot study. *BMJ open* **6** (2016).

33 Vincent, G. E. *et al.* The impact of breaking up prolonged sitting on glucose metabolism and cognitive function when sleep is restricted. *Neurobiology of sleep and circadian rhythms* **4**, 17-23 (2018).

34 Chrismas, B. C., Taylor, L., Cherif, A., Sayegh, S. & Bailey, D. P. Breaking up prolonged sitting with moderate-intensity walking improves attention and executive function in Qatari females. *PloS one* **14**, e0219565 (2019).

35 Stoner, L. *et al.* Effects of acute prolonged sitting on cerebral perfusion and executive function in young adults: A randomized cross-over trial. *Psychophysiology* **56**, e13457 (2019).

36 Bergouignan, A. *et al.* Effect of frequent interruptions of prolonged sitting on self-perceived levels of energy, mood, food cravings and cognitive function. *International Journal of Behavioral Nutrition and Physical Activity* **13**, 1-12 (2016).

37 Bailey, B. W. *et al.* The impact of exercise intensity on neurophysiological indices of food-related inhibitory control and cognitive control: A randomized crossover event-related potential (ERP) study. *NeuroImage* **237**, 118162 (2021).

For peer review only

38 Zimmer, P. *et al.* The effects of different aerobic exercise intensities on serum serotonin concentrations and their association with Stroop task performance: a randomized controlled trial. *European journal of applied physiology* **116**, 2025-2034 (2016).

39 Chang, Y., Labban, J., Gapin, J. & Etnier, J. The effects of acute exercise on cognitive performance: a meta-analysis. *Brain research* **1453**, 87-101 (2012).

40 Berridge, C. W. & Spencer, R. C. Differential cognitive actions of norepinephrine a2 and a1 receptor signaling in the prefrontal cortex. *Brain research* **1641**, 189-196 (2016).

41 American College of Sports Medicine. *ACSM's guidelines for exercise testing and prescription (10th ed.)*. (Lippincott williams & wilkins, 2018).

42 Aston-Jones, G. & Cohen, J. D. An integrative theory of locus coeruleus-norepinephrine function: adaptive gain and optimal performance. *Annu. Rev. Neurosci.* **28**, 403-450 (2005).

43 Gelbard-Sagiv, H., Magidov, E., Sharon, H., Hendler, T. & Nir, Y. Noradrenaline modulates visual perception and late visually evoked activity. *Current Biology* **28**, 2239-2249. e2236 (2018).

44 Dempsey, P. C. *et al.* Prolonged uninterrupted sitting increases fatigue in type 2 diabetes. *Diabetes research and clinical practice* **135**, 128-133 (2018).

45 Segal, S. K., Cotman, C. W. & Cahill, L. F. Exercise-induced noradrenergic activation enhances memory consolidation in both normal aging and patients with amnestic mild cognitive impairment. *Journal of Alzheimer's disease* **32**, 1011-1018 (2012).

46 Weinstein, G. *et al.* Glucose indices are associated with cognitive and structural brain measures in young adults. *Neurology* **84**, 2329-2337 (2015).

47 Xu, F. *et al.* Acute effect of glucose on cerebral blood flow, blood oxygenation, and oxidative metabolism. *Human brain mapping* **36**, 707-716 (2015).

48 Page, K. A. *et al.* Effects of fructose vs glucose on regional cerebral blood flow in brain regions involved with appetite and reward pathways. *Jama* **309**, 63-70 (2013).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

49 Marshall, R. S. *et al.* Recovery of brain function during induced cerebral hypoperfusion. *Brain* **124**, 1208-1217 (2001).

50 Peddie, M. C. *et al.* Breaking prolonged sitting reduces postprandial glycemia in healthy, normal-weight adults: a randomized crossover trial. *The American journal of clinical nutrition* **98**, 358-366 (2013).

51 Rosenthal, J. M. *et al.* The effect of acute hypoglycemia on brain function and activation: a functional magnetic resonance imaging study. *Diabetes* **50**, 1618-1626 (2001).

52 Rolfe, D. & Brown, G. C. Cellular energy utilization and molecular origin of standard metabolic rate in mammals. *Physiological reviews* **77**, 731-758 (1997).

53 Pontifex, M. B., Parks, A. C., Henning, D. A. & Kamijo, K. Single bouts of exercise selectively sustain attentional processes. *Psychophysiology* **52**, 618-625 (2015).

For peer review only

Breaking sitting and cognition

Table 1 Characteristics of the included studies

Author/Country	Participants	Design	Arms	Outcomes	Results
Wheeler et al. (2020) /Australia	N = 67 (female = 35) Age: ≥ 55 to ≤ 80 years (M ± SD: 67 ± 7 years) Weight status: overweight/obese (BMI: ≥ 25 to < 45 kg/m^2); MVPA < 150 min/week; sitting time > 5 h/day	Randomized cross-over	Trial started: ~8:00 #**1. SIT:** sitting (8 h). **2. EX+SIT:** sitting (1 h), continuous moderate-intensity (65–75% of the age-predicted HR_{max}) walking (30 min), sitting (6.5 h); total PA = 30 min. #**3. EX+BR:** sitting (1 h), continuous moderate-intensity (65–75% of the age-predicted HR_{max}) walking (30 min), sitting interrupted every 30 min with 3 min of light-intensity walking (3.2 km/h), sitting (6.5 h); total PA = 69 min.	1. Psychomotor function and processing speed (the detection test) 2. Attention (the identification test) 3.Visual learning (the one card learning test) 4. EFs (the Groton Maze Learning Test) 5.WM (one-back and two-back tests)	**Psychomotor function, attention, visual learning, and EFs:** EX + BR = SIT **WM:** EX+BR > SIT (ES = 0.25)
Stoner et al. (2019)/Hill	N = 20 (female = 14) Age: 18–35 years (M ± SD: 22 ± 3 years); Weight status: normal/overweight (BMI = 25.5 ± 6.1 kg/m^2); MVPA <	Randomized cross-over	Trial started: 6:00–10:00 #**1. SIT:** sitting (3 h). #**2. BR:** sitting interrupted with 10 calf raises at the 15- and 20-min points, and then every 10 min during the	EF-IC (the Stroop test)	**EF-IC:** BR < SIT (ES = -0.85)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

	150 min/week; sitting time (M ± SD: 8.2 ± 2.2 h/day)		following sitting time; total PA = ~9 min.		
Chrismas et al. (2019)/UK	N = 17 (female = 17) Age: 21–24 years (M = NR) Weight status and MVPA: NR; sitting time > 7 h/day	Randomized cross-over	Trial started: ~8:00 #**1. SIT:** sitting (5 h). #**2. BR:** sitting interrupted every 30 min with 3 min of moderate-intensity walking (6.0 km/h, RPE = 12–14); total PA = 27 min.	1. Attention and vigilance (rapid visual information processing and choice reaction time) 2. EF-IC (the Stroop test) 3. WM (serial 3's and 7's subtraction tasks)	**Attention and vigilance:** BR > SIT (ES = 0.89) **EF-IC:** BR > SIT (ES = 0.75) **EF-WM**: BR = SIT
Vincent et al (2018) /Australia	N = 6 (female = 0) Age: 20–35 years (M ± SD: 27 ± 4 years) Weight status: lean (waist circumference = ~84 ± 7 cm); MVPA and sitting time: NR	Randomized cross-over	#**1. SIT:** sitting (7 h); #**2. BR:** sitting interrupted every 30 min with 3 min of light-intensity walking (3.2 km/h); total PA = 51 min.	WM (the digit symbol substitution test)	**No significant effects**
Mullane, Buman, Zeigler, Crespo and Gaesser (2017)/USA	N = 9 (female = 7) Age: 18–58 years (M ± SD: 30 ± 15 years) Weight status: overweight/obese (BMI = 29 ± 3 kg/m^2); MVPA < 150 min/week; sitting time: NR	Randomized cross-over	Trial started: 8:00 #**1. SIT:** sitting (8 h). **2. STAND:** standing #**3. CYCLE:** cycling (work rate = 20 W, cadence = 25–30 rpm). #**4. WALK:** walking (1.6 km/h)	1. Psychomotor function (the detection test) 2. WM (the one-back test) 3. EFs (the set-shifting test)	**Psychomotor function:** CYCLE > SIT (ES = 1.1) WALK > SIT (ES = 0.42) CYCLE > WALK (ES = 0.63) **EF-WM:**

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

			STAND, CYCLE, and WALK for a predetermined time hourly: 10 min at 08:50 and 09:50, 15 min at 10:45 and 11:45, 20 min at 12:40 and 13:20, and 30 min at 14:00 and 15:30; total PA = 150 min.		CYCLE > SIT (ES = 0.91) WALK > SIT (ES = 0.74) **EF-CF:** CYCLE > SIT (ES = 0.78) WALK > SIT (ES = 0.66)
Bergouignan et al. (2016)/USA	N = 30 (female = 9) Age: 25–50 years (M ± SD: 31 ± 6 years) Weight status: non-obese (BMI = 18.5–29.9 kg/m^2); MVPA < 150 min/week; sitting time > 9 h/day	Randomized cross-over	Trial started: 8:00 #**1. SIT:** sitting (6 h). **2. EX:** continuous moderate-intensity (RPE = 12–13) walking (30 min); total PA = 30 min. #**3. BR:** sitting interrupted every 1 h with 5 min of moderate-intensity walking (RPE = 12–13); total PA = 30 min.	EF (Flanker task and trail-making test)	**No significant effects**
Wennberg et al. (2016)/Australia	N = 19 (female = 9) Age: 45–75 years (M ± SD: 60 ± 8 years) Weight status: overweight/obese (BMI: ≥ 25	Randomized cross-over	Trial started: 7:00–8:00 #**1. SIT:** sitting (7 h). #**2. BR:** sitting interrupted every 30 min with 3 min of light-intensity	1. EFs (the Flanker task and Stroop test) 2. Episodic memory (a face-name association test)	**No significant effects**

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	to ≤ 45 kg/m^2); MVPA < 150 min/week; sitting time > 5 h/day		walking (3.2 km/h, RPE = 9); total PA = 30 min.	3. WM (the n-back task and letter memory test)	

Abbreviations: M = mean; MVPA = moderate-to-vigorous intensity physical activity; SIT = sitting; EX = continuous exercise; BR = breaking sitting; HR_{max} = maximal heart rate; PA = physical activity; BMI = body mass index; CYCLE = cycling; NR = not reported; RPE = Rating of perceived exertion (Borg scale); EF = executive function; IC = inhibitory control; WM = working memory; [#] = included arms for review; In the result column: > = positive effect of interrupting sitting ($p < 0.05$); < = negative effect of interrupting sitting ($p < 0.05$); ES = effect size (standardized mean difference)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Table 2 Quality assessment of the included studies

Study	Eligibility criteria and source	Random allocation	Concealed allocation	Baseline comparability	Blinding of subjects	Blinding of therapists	Blinding of assessors	Completeness of follow-up	Intention -to-treat	Between-group comparison	Point and variability measures	Total scores
Wheeler et al. (2020)*	Yes	Yes	Yes	No	No	No	No	Yes	No	Yes	Yes	5
Stoner et al. (2019)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Chrismas et al. (2019)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Mullane et al. (2017)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Vincent et al. (2018)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Bergouignan et al. (2016)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Wennberg et al. (2016)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Mean score (SD)	4.1 (0.4)											

PEDro scoring: Yes = 1, No = 0, NR = 0; Item 1 (eligibility criteria and source) is not used to calculate the PEDro score; the following cut-points were used to describe the methodological quality of the studies: 9–10 (excellent), 6–8 (good), 4–5 (fair), and ≤ 3 (poor)

* = scores obtained from the PEDro database.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

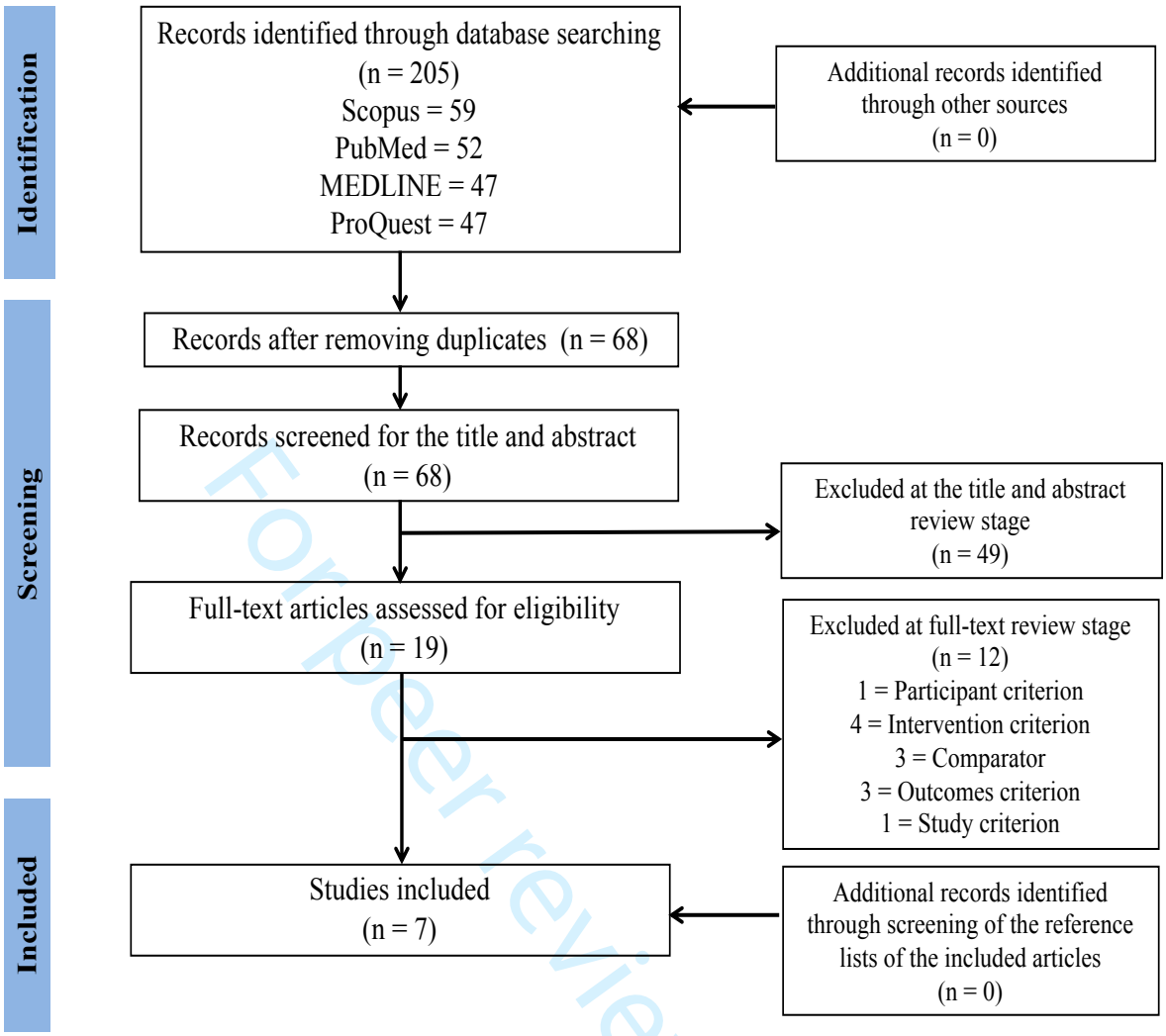


Figure 1. Flow diagram of the selection process showing the identification, screening, eligibility, and inclusion of studies.

For peer review only

Supplementary Appendix S1

Scopus	PubMed	MEDLINE	ProQuest
(TITLE-ABS ("Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting") AND TITLE-ABS (cognitive OR cognition OR executive OR brain OR cerebral))	("Prolonged sitting" [Title/Abstract] OR "Interrupted sitting" [Title/Abstract] OR "breaking sitting" [Title/Abstract] OR "uninterrupted sitting" [Title/Abstract] OR "continuous sitting"[Title/Abstract]) AND (cognitive [Title/Abstract] OR cognition [Title/Abstract] OR executive [Title/Abstract] OR brain [Title/Abstract] OR cerebral[Title/Abstract])	(TI ("Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting") AND TI (cognitive OR cognition OR executive OR brain OR cerebral)) OR (AB ("Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting") AND AB (cognitive OR cognition OR executive OR brain OR cerebral))	(ti("Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting") AND ti(cognitive OR cognition OR executive OR brain OR cerebral)) OR (ab("Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting") AND ab(cognitive OR cognition OR executive OR brain OR cerebral))

For peer review only



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	See title page
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	p.2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	p.1 to 2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	p.2
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	p.2
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	p.3
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	p.3
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	p.3
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	p.5
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	p.5
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	p.5
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	p.5
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NR
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	p.5
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	p.5
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	p.5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	NR
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NR
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NR
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NR

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NR
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	p.6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	p.4
Study characteristics	17	Cite each included study and present its characteristics.	p.6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	p.8
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	p.7
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	NR
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NR
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NR
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NR
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NR
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NR
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	p.8 to 9
	23b	Discuss any limitations of the evidence included in the review.	p.14
	23c	Discuss any limitations of the review processes used.	p.14
	23d	Discuss implications of the results for practice, policy, and future research.	p.14
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	p.2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	p.2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	p.2
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	p.15
Competing interests	26	Declare any competing interests of review authors.	p.15
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	p.3

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2020 Checklist

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

For peer review only

Title: The Acute Effect of Breaking up Prolonged Sitting on Cognition: A Systematic Review

Running head: Breaks from sitting and cognition

Word count: 4681

Authors:
Ting-Yu Chueh[1], Yung-Chih Chen*[1], Tsung-Min Hung*[1, 2]

Affiliations:
[1] Department of Physical Education and Sport Sciences, National Taiwan Normal University, Taipei, Taiwan
[2] Institute for Research Excellence and Learning Sciences, National Taiwan Normal University, Taipei, Taiwan

***Correspondence:**
Yung-Chih Chen, Ph.D.
E-mail: yc.chen@ntnu.edu.tw
Department of Physical Education and Sport Sciences, National Taiwan Normal University, Taipei, Taiwan
162, Section 1, Heping E. Rd., Taipei City 106, Taiwan

Tsung-Min Hung, Ph.D., FNAK, FISSP
E-mail: ernesthungkimo@yahoo.com.tw
Department of Physical Education and Sport Sciences & Institute for Research Excellence and Learning Sciences, National Taiwan Normal University, Taipei, Taiwan
162, Section 1, Heping E. Rd., Taipei City 106, Taiwan

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Abstract

Objectives: To review the current evidence on the acute effects of interrupting prolonged periods of sitting with intermittent physical activity (PA) on cognition in healthy populations.

Design: This systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.

Methods: Studies were included if they investigated the acute effects of taking regular PA breaks from sitting on cognition in healthy populations without any cardiovascular disease, history of brain injury, or psychiatric or neurological disorder. Four electronic databases—PubMed, Scopus, MEDLINE, and ProQuest—were searched for eligible studies on September 20, 2020. Study quality was assessed using the Physiotherapy Evidence Database scale. This review is registered in PROSPERO (number: CRD42020147536).

Results: Seven studies, involving 168 participants aged between 18 and 80 years, were eligible for inclusion in this review. Three of the seven studies found positive effects of interrupting sitting with either a) 3 min of relatively high-intensity (6 km/h) walking every 30 min on attention and inhibitory control in young adults; b) hourly breaks with progressively longer durations (10–30 min) of very light-intensity cycling/walking on attention, working memory, and cognitive flexibility in adults with obesity; or c) an initial bout of continuous moderate-intensity exercise, followed by interruption of post-exercise sitting with 3-min breaks of light-intensity walking (3.2 km/h) every 30 min, on working memory in older adults with overweight.

Conclusion: The small number of available studies provided preliminary results that interrupting periods of sitting may beneficially affect cognition in healthy adults. Given the limited evidence on this topic in the literature and the heterogeneity of PA protocols across the included studies, our results regarding the effectiveness of interrupting prolonged sitting with PA breaks in improving cognition warrant further verification.

Funding: Higher Education Sprout Project by Ministry of Education. Award/Grant number is not applicable.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Key words: Sedentary behavior; Cognition; Physical activity; Breaking sitting; Brain health

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and limitations of this study

- ✓ This is the first study to systematically review the acute effects of taking regular physical activity breaks from sitting on cognition in healthy individuals.
- ✓ This review highlights significant research gaps in the field.
- ✓ This review identifies potential mechanisms underlying the acute effects of taking physical activity breaks from sitting on cognition.
- ✓ A meta-analysis is not possible due to the methodological heterogeneity of the included studies.
- ✓ Firm conclusions are limited by the small number of available studies.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Introduction

Optimal cognitive performance in all cognitive domains, including attention, executive functions (EFs), and memory engagement, is important for optimizing daily functioning and reducing cognitive decline, which are critical to productivity and quality of life [1,2]. In classrooms and offices, for example, individuals require attention to focus on relevant information while relying on distinct EFs to suppress irrelevant distraction (inhibitory control), hold and mentally organize information (working memory [WM]), and adjust their behavior or thoughts based on updated demands, rules, or priorities (cognitive flexibility) to achieve goals [3-6]. These abilities have profound attribution effects on academic attainment and job performance [7,8]. In addition, a healthy cognitive state is beneficial in late life, because it reduces age-related cognitive decline [9]. Therefore, it is important to maintain and/or enhance cognition in daily life.

Sedentary behavior and physical activity (PA) are two important modifiable factors associated with cognition [10,11]. Adults typically spend the vast majority (8–10 h on average) of their waking day in sedentary behavior such as sitting and/or reclining [12,13]. Recent studies have indicated that increased sedentary behavior was associated with lower cognitive performance [14-16]. This association was partially independent of the PA level[17]. Studies also have demonstrated that prolonged sitting transiently deteriorated cognitive performance (i.e., increases problem solving and attention task errors) [18,19]. Although it has been well established that cognitive performance is enhanced after 20 to 30 min of continuous/structured exercise [20,21], individuals who engage in such exercise may be largely sedentary throughout the rest of the day [22]. Accordingly, taking regular PA breaks from sitting throughout the day may be a feasible strategy to offset the negative effects of prolonged sitting on cognition.

Taking regular PA breaks from sitting has been shown to be a practical approach to reduce the negative effects of acute prolonged sitting on cardiovascular and metabolic functions [23-25].

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Given that acute uninterrupted sitting has negative effects on cognition [18,19], related studies have extended the investigation from the physiological outcomes to the cognitive outcomes of taking regular breaks from sitting. Although recent studies have investigated the effects of taking breaks from sitting on cognition, they have not yet been subjected to a systematic review, which can provide a high level of confidence in the current findings. Furthermore, understanding the physiological mechanisms behind the acute effects of taking PA breaks from sitting on cognition is important to facilitate the design of successful PA approaches. Thus, the aims of this paper were (1) to systematically review studies investigating the acute effects of interrupting prolonged periods of sitting on cognition compared with uninterrupted sitting; (2) to provide a methodological assessment of all of the reviewed studies; and (3) to discuss the potential underlying physiological mechanisms of any effects found and provide suggestions for future studies.

Methods

This review was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines [26,27] and is registered in the International Prospective Register of Systematic Reviews (PROSPERO; identification code: CRD42020147536) [1].

Eligibility criteria

The Population, Intervention, Comparison, Outcome, and Study (PICOS) framework was used to determine the inclusion criteria for studies [26], as follows. (P) Participants: participants

[1] This study was initially planned to include studies that examined the acute effects of interrupting sitting on cerebrovascular function. However, only three studies met the eligibility criteria in the final search, and two of them involved the same dataset of cognition studies. Thus, we changed the focus of this systematic review to studies that evaluated the acute effects of interrupting sitting on cognition.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

were healthy and without any cardiovascular disease, history of brain injury, or psychiatric or neurological disorder. (I) Intervention: interventions were of an acute nature, involving interrupting prolonged periods of sitting with a regular pattern of PA at predefined intervals (e.g., every 30 min). (C) Comparator: studies included a non-treatment control group (i.e., the prolonged sitting condition). (O) Outcome: studies tested cognitive performance. (S) Study design: studies were randomized controlled or non-randomized controlled trials. Studies were excluded if they (1) involved only a single bout of PA in the intervention condition; (2) did not explicitly state the PA-related parameters, such as type, intensity, frequency, and duration; and/or (3) were not published in an English language peer-reviewed journal.

Information sources and search strategy

Four electronic databases—PubMed, Scopus, MEDLINE, and ProQuest—were searched for eligible studies from inception. The initial search was conducted on August 5, 2019, and the final search was completed on September 20, 2020. The key words used to search the titles and abstracts were discussed by the research team to maximize the chance of identifying relevant articles. The following key words were used: (*"Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting"*) *AND* (*cognitive OR cognition OR executive OR brain OR cerebral*). The full search terms used in all four databases are provided in Supplementary Appendix S1.

The inclusion and exclusion of articles were decided according to the PICOS criteria, with the screening and selection of studies being completed by two authors ([redacted]). First, the titles and abstracts were independently assessed by these two authors and initially coded as "yes," "no," or "maybe" for inclusion. The same two authors then reviewed the full texts of the "yes" and "maybe" studies, and disagreements regarding the inclusion of any study were resolved by discussion with a third reviewer ([redacted]). The reference lists of all of the included

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

articles were then searched to check for potentially relevant studies. **Figure 1** provides an overview of the selection process.

Data collection and items

The data collection was conducted by the same two authors ([redacted]). The authors thoroughly read the included studies and extracted the following data: (1) first author's name, publication year, and country of data collection; (2) participants' characteristics (viz., sample size, age, weight status, duration of exercise, and daily sedentary behavior); (3) study design; (4) details of the intervention protocols; and (5) outcome measures.

Synthesis methods

The results in this review were analyzed through a process of narrative synthesis. The data were first analyzed by one reviewer ([redacted]) and then verified by a second reviewer ([redacted]). As all of the necessary information could be obtained from the articles, no authors were contacted for information. A p-value of < 0.05, presented in the original studies, was used across the studies to determine the significant effects of interrupting sitting with PA on cognition. The results were converted to effect sizes (standardized mean difference) and entered into the results column of **Table 1**. Cohen's d was used to interpret the effect sizes: small ($d = 0.2$), medium ($d = 0.5$), and large ($d = 0.8$) [28].

Quality assessment

The quality of the included studies was assessed using the Physiotherapy Evidence Database (PEDro) scale [29], which produces a quantitative assessment of bias as a score out of 10, with higher scores indicating higher methodological quality. The PEDro scale assesses allocation bias (randomization, concealment of allocation), performance bias (blinding of participants and personnel), and detection bias (blinding of outcome assessment), in addition

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

to other items, including eligibility criteria, baseline comparability, retention rate, intention-to-treat, between-group statistical comparisons for at least one key outcome, point measures, and variability measures. When available, the PEDro score was obtained from the PEDro database. This was possible only for one study [30]. For studies for which the PEDro score was not available from the database, it was calculated by the two review authors ([redacted]) independently. Differences were resolved by discussion and adjudicated by a third author ([redacted]) as required.

Patient and public involvement

No patient involved.

Results

Study selection

In total, seven studies met all of the inclusion criteria and were included in this review. **Figure 1** illustrates the study selection process in a PRISMA flow diagram [26].

Study characteristics

All of the seven included studies were randomized crossover trials and comprised a total of 91 male and 77 female participants aged between 18 and 80 years. The body mass index (BMI) of the participants across all of the studies was between 18.5 and 44 kg/m^2. Most of the participants (57%) were overweight or obese (BMI > 25 kg/m^2) [30-32], and most (86%) did not report being actively engaged in regular exercise, defined as engaging in moderate to vigorous physical activity (MVPA) for at least 150 min per week. The duration of sedentary behavior across all studies was between 5 and 9 h per day. **Table 1** provides a summary of the study designs and participant characteristics.

Regarding the PA intensity during breaks from sitting, four out of the seven studies stated that the exercise was of light-intensity (walking on a treadmill at 1.6–3.6 km/h or cycling at 20

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

watts with 25–30 rpm) [30-33], two studies reported that the break consisted of moderate-intensity PA (walking on a treadmill at ~ 6 km/h) [34] and one study implemented 10 calf raises at a rate of 0.33 Hz (20 per minute) [35].

With regard to the duration and frequency of breaks, three studies implemented 3-min PA breaks from sitting every 30 min, for a total daily exercise duration of 27 [34], 30 [32], and 51 min [33]. In one study, every 60 min of sitting was interrupted with 5 min of PA, for a total daily PA duration of 30 min [36], and in another study, every 60 min of sitting was interrupted with progressively longer durations of either walking or cycling (i.e., 10, 15, 20, and 30 min), for a daily total of 150 min of very low-intensity PA [31]. One study instructed the participants to performed 10 calf raises within 30 s after every 5 or 10 min of sitting, for a total daily PA duration of 9 min (17 sets of calf exercise) [35]. The remaining study included an initial 30-min bout of moderate-intensity walking, followed by sitting interrupted every 30 min with 3 min of PA, for a total daily PA duration of 69 min [30]. **Table 1** summarizes the intervention characteristics for all of the included studies.

Results of individual studies

All of the included studies focused on EFs, three studies evaluated attention or processing speed [30,31,34], two assessed memory [31], and one study used a visual learning task [30]. Three studies found positive effects of PA breaks from sitting on EFs. In one study, compared with prolonged sitting, progressively longer durations of both light-intensity walking and cycling (i.e., 10, 15, 20, and 30 min) after every 60 min of sitting resulted in greater attention, WM, and cognitive flexibility, with small to large effect sizes (d = 0.62–0.91) [31]. Interestingly, cycling resulted in a better improvement in attention than walking, with a medium effect size (d = 0.63). Taking 3-min breaks of moderate-intensity PA (6.0 km/h, rating of perceived exertion [RPE] = 12–14) every 30 min resulted in improved inhibitory control with a large effect size (d = 0.74) [34], whereas taking 3-min breaks of light-intensity walking (3.2 km/h)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

every 30 min [32,33] or 5-min breaks of moderate-intensity walking (RPE = 12–13) every 60 min showed no significant effect on any cognitive performance measure [36]. Three-minute breaks of light-intensity PA (3.2 km/h) every 30 min after an initial bout of continuous moderate-intensity exercise (65–75% of the age-predicted maximal heart rate) had a beneficial effect on WM (d = 0.25) [30]. One study found adverse effects of intermittent 10 calf raise exercises on inhibitory control (d = -0.85) [35]. Two studies reported the positive effects of taking 3-min breaks of moderate-intensity PA every 30 min or progressively longer breaks of light-intensity walking and cycling every 60 min on attention, with medium to large effect sizes (d = 0.42 to 1.1) [31,34]. Interestingly, cycling resulted in a better improvement in attention than walking, with a medium effect size (d = 0.63). No significant effects of interrupting sitting with PA was found in other cognitive domains.

Methodological quality

The included studies had a mean PEDro score of 4.1, indicating "fair" methodological quality. All studies reported to have performed random allocation of participants to the trials, but only one study reported performing concealed allocation [30]. This suggests the presence of allocation biases in the other studies. None of the studies mentioned the blinding of participants and/or researchers because it is impossible to blind an intervention that requires the participants to perform activities. All of the studies provided a key outcome measure for more than 85% of their participants, and all of the studies adopted a crossover trial design, which meant that all participants received all of the designed treatments. Nevertheless, none of the studies explicitly stated whether at least one main outcome was included in the statistical analysis (intention-to-treat analysis) of all of the participants, regardless of any subsequent withdrawal from treatment. This suggests the presence of biased comparisons between the treatment arms. Finally, all of the studies provided statistical comparisons and valid measures for at least one key outcome measure of interest. Detailed results of the PEDro assessment are presented in **Table 2**.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Discussion

The purpose of this study was to review the results of studies that investigated whether interrupting prolonged periods of sitting with PA breaks would improve cognition. Seven studies were identified that investigated the acute effects of such interruptions in healthy individuals. All of these studies were published recently (between 2016 and 2020). Three of the seven studies provided preliminary evidence that taking regular PA breaks from sitting offers short-term benefits for cognition in healthy adults. However, no definite conclusions could be drawn regarding the effectiveness of taking PA breaks from sitting in improving cognition due to the heterogeneity of the PA protocols across the included studies and the small number of available studies.

The acute effects of taking PA breaks from sitting on cognition

This review suggests that taking PA breaks from sitting, specifically their intensity and/or volume, might play a potential role in regulation of cognitive performance. For example, compared with prolonged sitting, taking 3-min breaks of moderate-intensity walking (6 km/h) every 30 min was reported to result in better attention and inhibitory control in young female adults [34]. However, the same protocol (3 min of PA every 30 min) but with light-intensity walking (3.2 km/h) produced no effect on episodic memory, inhibitory control, or WM in overweight/obese middle-aged adults [32] or on attention or WM in young adults [33]. Studies have reported that exercise intensity may be an important determinant of cognitive benefits [37,38]. A meta-analysis demonstrated that the positive acute effect of exercise on cognition is seen when the exercise intensity is moderate, but the effect diminishes when the exercise intensity is very light [39]. One possible explanation is that greater exercise intensity is needed to induce sufficient

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

levels of norepinephrine (NE) to facilitate cognitive performance [40]. Thus, the intensity of PA during breaks from sitting may be one of the factors determining the effectiveness of the breaks.

However, taking higher-intensity PA breaks is not always beneficial if the frequency of such breaks is low. For example, another study of non-obese adults investigated the effects of a similar walking speed (~5.7 km/h), but less frequent (hourly) and slightly longer (5 min), walking breaks from sitting and found no effect on the measures of inhibitory control or cognitive flexibility [36]. This implies that the benefits of taking relatively high intensity PA breaks from sitting might be reduced by having infrequent breaks. Interestingly, taking hourly low-intensity PA breaks might still benefit cognition if these breaks are sufficiently long (i.e., 10, 15, 20, and 30 min). As indicated in one study [31], both very low-intensity cycling (work rate = 20 W, cadence = 25–30 rpm) and walking (1.6 km/h) resulted in better attention (the detection test), cognitive flexibility (the set-shifting test), and WM (the one-back test) compared with prolonged sitting. Despite the low intensity of PA breaks, the longer duration of PA breaks adopted in this study, which resulted in a greater amount of exercise (150 min in total), might have had a compensating effect. This finding implies that volume of exercise should be considered and carries important implication for some individuals who are unable to engage in moderate to vigorous intensity exercise. Collectively, these results suggest that the effects of taking breaks from sitting on cognitive function are dependent on the intensity, frequency, and duration of the PA performed during the breaks. Future studies are recommended to determine the minimum number and duration of low-intensity PA breaks from sitting required for a meaningful improvement in cognition.

The study that evaluated the effects of a morning bout of 30 min of moderate-intensity exercise with or without subsequent light-intensity PA breaks every 30 min on cognitive performance in obese/overweight older adults provides another fruitful line of research [30]. A single bout of moderate-intensity PA before engaging in prolonged sitting showed better EFs, but not WM, relative to the prolonged sitting condition, whereas the interruption of post-

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

exercise sitting with several additional bouts of light-intensity PA led to improved WM, but not EFs [30]. These results suggest that a single bout of continuous moderate-intensity exercise with or without subsequent PA breaks provide benefits in different cognitive domains. These divergent effects may be attributed to individual differences in the level of aerobic fitness and the timing and order of the administration of cognitive tests following the cessation of exercise. It is reasonable to assume that the participants in that study had lower aerobic fitness, based on the provided demographic data (old age, sedentary lifestyle, and obese/overweight status). In this study, the test of EFs was administered almost immediately (approximately 2 min) after the 3-min light-intensity PA breaks, followed by the WM test and other cognitive assessments. A previous meta-analytic review indicated that the largest effect of lower-intensity PA on cognitive performance was immediately after the PA, whereas that of higher-intensity activities was after a delay of ~10 min [39]. However, an individual's level of aerobic fitness may moderate the physiological demands and/or perception of the physical activity stimulus [41]. Chang, et al. [39] observed that aerobic fitness was a moderating variable when cognition was assessed immediately after the cessation of a bout of PA. Thus, the beneficial effects of a single bout of continuous moderate-intensity exercise with subsequent light-intensity PA breaks on EFs in this study may have been diminished due to the lower aerobic fitness level of the participants and the small time interval between exercise cessation and EF assessment. Moreover, when the aim is to maintain WM, it appears that it is necessary to avoid prolonged sitting even after an exercise session.

Of note, one study found adverse effects of frequent calf raise exercises on inhibitory control in young adults [35]. The impaired inhibitory control performance was accompanied by a decreased prefrontal cortex perfusion, as measured by near-infrared spectroscopy, supporting that prefrontal cortex activity could be a potential mechanism for modulating cognitive performance [6]. The negative effect of acute exercise on EFs was observed only after a bout of high-intensity exercise [39]. We found no study reporting a negative effect on cognition following

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

a short bout of low-intensity exercise, such as calf raises. The reason for this result is unknown; therefore, future studies should attempt to replicate this result and investigate the reason.

Taken together, the literature reviewed suggests that to understand the effects of PA breaks from prolonged sitting on cognitive performance, the effects of the frequency, intensity, and duration of the PA undertaken during the breaks need to be considered concurrently. The current evidence suggests that a) short, frequent breaks (at least every 30 min) involving moderate-intensity PA (walking at least 6 km/h); b) longer, hourly breaks involving even very low-intensity PA; and c) a single bout of 30-min continuous moderate-intensity exercise with subsequent short PA breaks are all beneficial to cognitive performance (e.g., attention, EFs, or memory-related tasks). However, the heterogeneity of PA protocols across the reviewed studies and the small number of available studies make direct comparisons of study results difficult. Furthermore, the findings of the effectiveness of PA breaks from sitting in improving cognition need to be further confirmed, because fewer than half of the studies included reported significant positive results.

Potential mechanisms underlying the effects of interrupting sitting on cognition

Despite the inconsistent results across the reviewed studies, the available evidence suggests that taking PA breaks from prolonged sitting exerts acute beneficial effects on cognition. Several potential mechanisms may explain why interrupting periods of sitting is beneficial to brain function.

First, research in both humans and animal models suggests that the activation of the locus coeruleus and the associated release of NE play important roles in influencing the attention state, which may modulate cognitive performance [42,43]. NE release is at least partly modulated by the amount of PA. Studies have found that prolonged uninterrupted sitting is associated with increased levels of fatigue [44] and decreased plasma levels of NE [32], whereas moderate-intensity exercise can induce NE release (indexed by salivary alpha-amylase concentrations) and

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

enhance cognitive performance [32,45]. Two studies that used either frequent breaks of relatively light-intensity PA (every 30 min) or less frequent breaks of moderate-intensity PA (every 60 min) found no beneficial effects on the plasma and urinary levels of NE or cognitive performance [32,36]. Moderate-intensity PA breaks every 30 min; progressively increasing the duration of breaks, even at light-intensity PA; and performing an initial bout of continuous moderate-intensity exercise for 30 min before subsequent light-intensity PA breaks during sitting, all resulted in improved cognitive performance compared with the prolonged sitting condition [31,34]. It is known that cognitive performance is enhanced with increasing levels of NE [40], which, in turn, is affected by PA. Thus, individuals interested in taking regular physical activity breaks from sitting for cognitive benefits should take into account the combination of intensity, frequency, and duration.

Secondly, glucose plays a vital role in supplying energy to cranial nerves and so glucose metabolism may influence cognition. The glucose-centric hypothesis postulates that exposure to acute hyperglycemia impairs cognition [11]. Indeed, young and middle-aged people with diabetes have shown impaired performance on measures of memory, visual perception, and attention. This may be partly due to brain injury through neurodegeneration, frontal and hippocampal atrophy, and injury to the white matter microstructure [46]. Studies have demonstrated that acute glucose ingestion (i.e., blood glucose concentrations increasing from 90 to 142 mg/dl) reduces the global cerebral metabolic rate of oxygen [47] and regional cerebral blood flow (CBF) [48], which result in poorer cognitive performance [49]. Regularly interrupting periods of sitting has been shown to reduce abrupt elevations in postprandial glucose concentrations [23,25]. One study found that in healthy, normal-weight adults, taking short but frequent PA breaks (every 30 min) resulted in higher CBF than taking longer but less frequent PA breaks (every 120 min). This is in line with the results of a previous study that examined the acute effects of taking regular PA breaks with different bouts of PA on blood glucose [50]. These findings suggest that regularly taking PA breaks after meals might be beneficial to

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

cognition by preventing both the reductions in cerebral oxygen and regional CBF and repeated subtle brain injuries. On the other hand, it has been shown that hypoglycemia (from 100 to 48 mg/dl) also acutely reduces blood oxygen level-dependent cortical activation and cognitive function [51], as the brain requires a high level of energy supply [52]. Therefore, to avoid hyperinsulinemic hypoglycemia after meal consumption, taking breaks from sitting may be a useful means of maintaining blood glucose concentration within a stable range. In summary, studies suggest that taking breaks from sitting might benefit cognition by means of affecting glucose metabolism.

Limitations and suggestions for future research

Some caution is warranted when interpreting the findings of this review. First, the results cannot be generalised beyond the specific populations studied, i.e. children, adolescents, and individuals with chronic disease such as type 2 diabetes. The current literature has not focused on school-aged children. School-aged children spend the majority of class time sitting and this might negatively affect their cognitive function (i.e., attention state) [53]. In addition, individuals with type 2 diabetes showed impaired glucose metabolism that might, according to the glucose-centric hypothesis, modulate the relationship between PA and cerebrovascular function [11]. Thus, these populations need to be independently examined. Second, given the heterogeneity of PA protocols, cognitive assessments, and sample size across the included studies, more studies with larger samples are needed to clarify the role of different PA characteristics (i.e., frequency, intensity, duration, and type) in the effect of PA breaks from sitting on distinct cognitive domains so as to provide more precise recommendations for public health. Third, this review exclusively focused on studies that assessed the acute effects of PA breaks on cognition. Future research should explore the accumulative effects of multiple days of sitting with or without PA breaks on cognition. Fourth, there is a need to further explore the underlying mechanisms using biochemical and physiological measures, such as neural activity in the locus coeruleus NE

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

system and glucose metabolism. Finally, future studies should examine the isolated contributions of interrupting prolonged sitting versus continuous PA to improvements in cognition.

Conclusions

The current review provides preliminary evidence that the intensity and/or volume of PA breaks from prolonged sitting may determine the effects of such breaks on cognition. However, due to the heterogeneity of PA protocols across studies and limited evidence, firm conclusions regarding the effectiveness of interrupting sitting in improving cognition cannot be drawn. Further studies are warranted to examine the duration, frequency, and intensity of PA breaks required to optimize cognitive performance.

Conflict of interest: There is no conflict of interest to report.

Ethics approval: Ethical approval was not applicable because this study retrieved data from already published studies.

Author contributions: TYC conceived the study concept; contributed to the study design, literature search, and data screening and extraction; assessed the article quality; and wrote the first draft of the manuscript. YCC conceived the study concept; contributed to the study design, literature search, and data screening and extraction; assessed the article quality; and critically reviewed and revised the manuscript. TMH conceived the study concept, contributed to the study design, assisted with data screening and extraction and quality assessment, and critically reviewed and revised the manuscript. All of the authors approved the submission.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

357

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

References

1 Hill, N. L. *et al.* Subjective cognitive impairment and quality of life: a systematic review. *International Psychogeriatrics* **29**, 1965-1977 (2017).

2 Sanz, J. H. *et al.* Executive function and psychosocial quality of life in school age children with congenital heart disease. *J Pediatr* **202**, 63-69 (2018).

3 Diamond, A. Executive functions. *Annu Rev Psychol* **64**, 135-168 (2013).

4 Miyake, A. *et al.* The unity and diversity of executive functions and their contributions to complex "frontal lobe" tasks: A latent variable analysis. *Cognitive psychology* **41**, 49-100 (2000).

5 Zelazo, P. D. Executive function: Reflection, iterative reprocessing, complexity, and the developing brain. *Developmental Review* **38**, 55-68 (2015).

6 Miller, E. K. & Cohen, J. D. An integrative theory of prefrontal cortex function. *Annual review of neuroscience* **24**, 167-202 (2001).

7 Lang, J. W., Kersting, M., Hülsheger, U. R. & Lang, J. General mental ability, narrower cognitive abilities, and job performance: The perspective of the nested-factors model of cognitive abilities. *Personnel psychology* **63**, 595-640 (2010).

8 Rohde, T. E. & Thompson, L. A. Predicting academic achievement with cognitive ability. *Intelligence* **35**, 83-92 (2007).

9 Buckner, R. L. Memory and executive function in aging and AD: multiple factors that cause decline and reserve factors that compensate. *Neuron* **44**, 195-208 (2004).

10 Erickson, K. I. *et al.* Physical Activity, Cognition, and Brain Outcomes: A Review of the 2018 Physical Activity Guidelines. *Medicine and science in sports and exercise* **51**, 1242-1251 (2019).

11 Wheeler, M. J. *et al.* Sedentary behavior as a risk factor for cognitive decline? A focus on the influence of glycemic control in brain health. *Alzheimer's & Dementia: Translational Research & Clinical Interventions* **3**, 291-300 (2017).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12 Castro, O., Bennie, J., Vergeer, I., Bosselut, G. & Biddle, S. J. How sedentary are university students? A systematic review and meta-analysis. *Prevention Science* **21**, 332-343 (2020).

13 Van Dyck, D. *et al.* International study of objectively measured physical activity and sedentary time with body mass index and obesity: IPEN adult study. *International Journal of Obesity* **39**, 199-207 (2015).

14 Bakrania, K. *et al.* Associations between sedentary behaviors and cognitive function: Cross-sectional and prospective findings from the UK biobank. *American journal of epidemiology* **187**, 441-454 (2017).

15 Falck, R. S., Davis, J. C. & Liu-Ambrose, T. What is the association between sedentary behaviour and cognitive function? A systematic review. *Br J Sports Med* **51**, 800-811 (2017).

16 Vancampfort, D. *et al.* Mild cognitive impairment and sedentary behavior: a multinational study. *Experimental gerontology* **108**, 174-180 (2018).

17 Ku, P.-W., Liu, Y.-T., Lo, M.-K., Chen, L.-J. & Stubbs, B. Higher levels of objectively measured sedentary behavior is associated with worse cognitive ability: Two-year follow-up study in community-dwelling older adults. *Experimental gerontology* **99**, 110-114 (2017).

18 Triglav, J. *et al.* Physiological and cognitive measures during prolonged sitting: Comparisons between a standard and multi-axial office chair. *Applied ergonomics* **78**, 176-183 (2019).

19 Baker, R., Coenen, P., Howie, E., Williamson, A. & Straker, L. The short term musculoskeletal and cognitive effects of prolonged sitting during office computer work. *International journal of environmental research and public health* **15**, 1678 (2018).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

20 Oberste, M. *et al.* Effects and moderators of acute aerobic exercise on subsequent interference control: a systematic review and meta-analysis Running title: Acute exercise and interference control. *Frontiers in Psychology* **10**, 2616 (2019).

21 Ludyga, S., Gerber, M., Brand, S., Holsboer-Trachsler, E. & Pühse, U. Acute effects of moderate aerobic exercise on specific aspects of executive function in different age and fitness groups: A meta-analysis. *Psychophysiology* **53**, 1611-1626 (2016).

22 Dempsey, P. C., Owen, N., Biddle, S. J. & Dunstan, D. W. Managing sedentary behavior to reduce the risk of diabetes and cardiovascular disease. *Current diabetes reports* **14**, 522 (2014).

23 Chen, Y.-C., Betts, J. A., Walhin, J.-P. & Thompson, D. Adipose Tissue Responses to Breaking Sitting in Men and Women with Central Adiposity. *Medicine and science in sports and exercise* **50**, 2049-2057 (2018).

24 Dunstan, D. W. *et al.* Breaking up prolonged sitting reduces postprandial glucose and insulin responses. *Diabetes care* **35**, 976-983 (2012).

25 Saunders, T. J. *et al.* The acute metabolic and vascular impact of interrupting prolonged sitting: a systematic review and meta-analysis. *Sports medicine* **48**, 2347-2366 (2018).

26 Moher, D. *et al.* Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic reviews* **4**, 1 (2015).

27 Moher, D., Liberati, A., Tetzlaff, J. & Altman, D. G. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Int J Surg* **8**, 336-341 (2010).

28 Cohen, J. Statistical Power Analysis for the Behavioral Sciences–Second Edition. 12 Lawrence Erlbaum Associates Inc. *Hillsdale, New Jersey* **13** (1988).

29 Maher, C. G., Sherrington, C., Herbert, R. D., Moseley, A. M. & Elkins, M. Reliability of the PEDro scale for rating quality of randomized controlled trials. *Physical therapy* **83**, 713-721 (2003).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

30 Wheeler, M. J. *et al.* Distinct effects of acute exercise and breaks in sitting on working memory and executive function in older adults: a three-arm, randomised cross-over trial to evaluate the effects of exercise with and without breaks in sitting on cognition. *Br J Sports Med* **54**, 776-781 (2020).

31 Mullane, S. L., Buman, M. P., Zeigler, Z. S., Crespo, N. C. & Gaesser, G. A. Acute effects on cognitive performance following bouts of standing and light-intensity physical activity in a simulated workplace environment. *J Sci Med Sport* **20**, 489-493 (2017).

32 Wennberg, P. *et al.* Acute effects of breaking up prolonged sitting on fatigue and cognition: a pilot study. *BMJ open* **6** (2016).

33 Vincent, G. E. *et al.* The impact of breaking up prolonged sitting on glucose metabolism and cognitive function when sleep is restricted. *Neurobiology of sleep and circadian rhythms* **4**, 17-23 (2018).

34 Chrismas, B. C., Taylor, L., Cherif, A., Sayegh, S. & Bailey, D. P. Breaking up prolonged sitting with moderate-intensity walking improves attention and executive function in Qatari females. *PloS one* **14**, e0219565 (2019).

35 Stoner, L. *et al.* Effects of acute prolonged sitting on cerebral perfusion and executive function in young adults: A randomized cross-over trial. *Psychophysiology* **56**, e13457 (2019).

36 Bergouignan, A. *et al.* Effect of frequent interruptions of prolonged sitting on self-perceived levels of energy, mood, food cravings and cognitive function. *International Journal of Behavioral Nutrition and Physical Activity* **13**, 1-12 (2016).

37 Bailey, B. W. *et al.* The impact of exercise intensity on neurophysiological indices of food-related inhibitory control and cognitive control: A randomized crossover event-related potential (ERP) study. *NeuroImage* **237**, 118162 (2021).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

38 Zimmer, P. *et al.* The effects of different aerobic exercise intensities on serum serotonin concentrations and their association with Stroop task performance: a randomized controlled trial. *European journal of applied physiology* **116**, 2025-2034 (2016).

39 Chang, Y., Labban, J., Gapin, J. & Etnier, J. The effects of acute exercise on cognitive performance: a meta-analysis. *Brain research* **1453**, 87-101 (2012).

40 Berridge, C. W. & Spencer, R. C. Differential cognitive actions of norepinephrine a2 and a1 receptor signaling in the prefrontal cortex. *Brain research* **1641**, 189-196 (2016).

41 American College of Sports Medicine. *ACSM's guidelines for exercise testing and prescription (10th ed.)*. (Lippincott williams & wilkins, 2018).

42 Aston-Jones, G. & Cohen, J. D. An integrative theory of locus coeruleus-norepinephrine function: adaptive gain and optimal performance. *Annu. Rev. Neurosci.* **28**, 403-450 (2005).

43 Gelbard-Sagiv, H., Magidov, E., Sharon, H., Hendler, T. & Nir, Y. Noradrenaline modulates visual perception and late visually evoked activity. *Current Biology* **28**, 2239-2249. e2236 (2018).

44 Dempsey, P. C. *et al.* Prolonged uninterrupted sitting increases fatigue in type 2 diabetes. *Diabetes research and clinical practice* **135**, 128-133 (2018).

45 Segal, S. K., Cotman, C. W. & Cahill, L. F. Exercise-induced noradrenergic activation enhances memory consolidation in both normal aging and patients with amnestic mild cognitive impairment. *Journal of Alzheimer's disease* **32**, 1011-1018 (2012).

46 Weinstein, G. *et al.* Glucose indices are associated with cognitive and structural brain measures in young adults. *Neurology* **84**, 2329-2337 (2015).

47 Xu, F. *et al.* Acute effect of glucose on cerebral blood flow, blood oxygenation, and oxidative metabolism. *Human brain mapping* **36**, 707-716 (2015).

48 Page, K. A. *et al.* Effects of fructose vs glucose on regional cerebral blood flow in brain regions involved with appetite and reward pathways. *Jama* **309**, 63-70 (2013).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

49 Marshall, R. S. *et al.* Recovery of brain function during induced cerebral hypoperfusion. *Brain* **124**, 1208-1217 (2001).

50 Peddie, M. C. *et al.* Breaking prolonged sitting reduces postprandial glycemia in healthy, normal-weight adults: a randomized crossover trial. *The American journal of clinical nutrition* **98**, 358-366 (2013).

51 Rosenthal, J. M. *et al.* The effect of acute hypoglycemia on brain function and activation: a functional magnetic resonance imaging study. *Diabetes* **50**, 1618-1626 (2001).

52 Rolfe, D. & Brown, G. C. Cellular energy utilization and molecular origin of standard metabolic rate in mammals. *Physiological reviews* **77**, 731-758 (1997).

53 Pontifex, M. B., Parks, A. C., Henning, D. A. & Kamijo, K. Single bouts of exercise selectively sustain attentional processes. *Psychophysiology* **52**, 618-625 (2015).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Table 1 Characteristics of the included studies

Author/Country	Participants	Design	Arms	Outcomes	Results
Wheeler et al. (2020) /Australia	N = 67 (female = 35) Age: ≥ 55 to ≤ 80 years (M ± SD: 67 ± 7 years) Weight status: overweight/obese (BMI: ≥ 25 to < 45 kg/m^2); MVPA < 150 min/week; sitting time > 5 h/day	Randomized cross-over	Trial started: ~8:00 #**1. SIT:** sitting (8 h). **2. EX+SIT:** sitting (1 h), continuous moderate-intensity (65–75% of the age-predicted HR_{max}) walking (30 min), sitting (6.5 h); total PA = 30 min. #**3. EX+BR:** sitting (1 h), continuous moderate-intensity (65–75% of the age-predicted HR_{max}) walking (30 min), sitting interrupted every 30 min with 3 min of light-intensity walking (3.2 km/h), sitting (6.5 h); total PA = 69 min.	1. Psychomotor function and processing speed (the detection test) 2. Attention (the identification test) 3.Visual learning (the one card learning test) 4. EFs (the Groton Maze Learning Test) 5.WM (one-back and two-back tests)	**Psychomotor function, attention, visual learning, and EFs:** EX + BR = SIT **WM:** EX+BR > SIT (ES = 0.25)
Stoner et al. (2019)/Hill	N = 20 (female = 14) Age: 18–35 years (M ± SD: 22 ± 3 years); Weight status: normal/overweight (BMI = 25.5 ± 6.1 kg/m^2); MVPA <	Randomized cross-over	Trial started: 6:00–10:00 #**1. SIT:** sitting (3 h). #**2. BR:** sitting interrupted with 10 calf raises at the 15- and 20-min points, and then every 10 min during the	EF-IC (the Stroop test)	**EF-IC:** BR < SIT (ES = -0.85)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	150 min/week; sitting time (M ± SD: 8.2 ± 2.2 h/day)		following sitting time; total PA = ~9 min.		
Chrismas et al. (2019)/UK	N = 17 (female = 17) Age: 21–24 years (M = NR) Weight status and MVPA: NR; sitting time > 7 h/day	Randomized cross-over	Trial started: ~8:00 #**1. SIT:** sitting (5 h). #**2. BR:** sitting interrupted every 30 min with 3 min of moderate-intensity walking (6.0 km/h, RPE = 12–14); total PA = 27 min.	1. Attention and vigilance (rapid visual information processing and choice reaction time) 2. EF-IC (the Stroop test) 3. WM (serial 3's and 7's subtraction tasks)	**Attention and vigilance:** BR > SIT (ES = 0.89) **EF-IC:** BR > SIT (ES = 0.75) **EF-WM**: BR = SIT
Vincent et al (2018) /Australia	N = 6 (female = 0) Age: 20–35 years (M ± SD: 27 ± 4 years) Weight status: lean (waist circumference = ~84 ± 7 cm); MVPA and sitting time: NR	Randomized cross-over	#**1. SIT:** sitting (7 h); #**2. BR:** sitting interrupted every 30 min with 3 min of light-intensity walking (3.2 km/h); total PA = 51 min.	WM (the digit symbol substitution test)	**No significant effects**
Mullane, Buman, Zeigler, Crespo and Gaesser (2017)/USA	N = 9 (female = 7) Age: 18–58 years (M ± SD: 30 ± 15 years) Weight status: overweight/obese (BMI = 29 ± 3 kg/m^2); MVPA < 150 min/week; sitting time: NR	Randomized cross-over	Trial started: 8:00 #**1. SIT:** sitting (8 h). **2. STAND:** standing #**3. CYCLE:** cycling (work rate = 20 W, cadence = 25–30 rpm). #**4. WALK:** walking (1.6 km/h)	1. Psychomotor function (the detection test) 2. WM (the one-back test) 3. EFs (the set-shifting test)	**Psychomotor function:** CYCLE > SIT (ES = 1.1) WALK > SIT (ES = 0.42) CYCLE > WALK (ES = 0.63) **EF-WM:**

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

			STAND, CYCLE, and WALK for a predetermined time hourly: 10 min at 08:50 and 09:50, 15 min at 10:45 and 11:45, 20 min at 12:40 and 13:20, and 30 min at 14:00 and 15:30; total PA = 150 min.		CYCLE > SIT (ES = 0.91) WALK > SIT (ES = 0.74) **EF-CF:** CYCLE > SIT (ES = 0.78) WALK > SIT (ES = 0.66)
Bergouignan et al. (2016)/USA	N = 30 (female = 9) Age: 25–50 years (M ± SD: 31 ± 6 years) Weight status: non-obese (BMI = 18.5–29.9 kg/m^2); MVPA < 150 min/week; sitting time > 9 h/day	Randomized cross-over	Trial started: 8:00 #**1. SIT:** sitting (6 h). **2. EX:** continuous moderate-intensity (RPE = 12–13) walking (30 min); total PA = 30 min. #**3. BR:** sitting interrupted every 1 h with 5 min of moderate-intensity walking (RPE = 12–13); total PA = 30 min.	EF (Flanker task and trail-making test)	**No significant effects**
Wennberg et al. (2016)/Australia	N = 19 (female = 9) Age: 45–75 years (M ± SD: 60 ± 8 years) Weight status: overweight/obese (BMI: ≥ 25	Randomized cross-over	Trial started: 7:00–8:00 #**1. SIT:** sitting (7 h). #**2. BR:** sitting interrupted every 30 min with 3 min of light-intensity	1. EFs (the Flanker task and Stroop test) 2. Episodic memory (a face-name association test)	**No significant effects**

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

	to ≤ 45 kg/m^2); MVPA < 150 min/week; sitting time > 5 h/day		walking (3.2 km/h, RPE = 9); total PA = 30 min.	3. WM (the n-back task and letter memory test)	

Abbreviations: M = mean; MVPA = moderate-to-vigorous intensity physical activity; SIT = sitting; EX = continuous exercise; BR = breaking sitting; HR_{max} = maximal heart rate; PA = physical activity; BMI = body mass index; CYCLE = cycling; NR = not reported; RPE = Rating of perceived exertion (Borg scale); EF = executive function; IC = inhibitory control; WM = working memory; [#] = included arms for review; In the result column: > = positive effect of interrupting sitting ($p < 0.05$); < = negative effect of interrupting sitting ($p < 0.05$); ES = effect size (standardized mean difference)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Table 2 Quality assessment of the included studies

Study	Eligibility criteria and source	Random allocation	Concealed allocation	Baseline comparability	Blinding of subjects	Blinding of therapists	Blinding of assessors	Completeness of follow-up	Intention -to-treat	Between-group comparison	Point and variability measures	Total scores
Wheeler et al. (2020)*	Yes	Yes	Yes	No	No	No	No	Yes	No	Yes	Yes	5
Stoner et al. (2019)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Chrismas et al. (2019)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Mullane et al. (2017)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Vincent et al. (2018)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Bergouignan et al. (2016)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Wennberg et al. (2016)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Mean score (SD)	4.1 (0.4)											

PEDro scoring: Yes = 1, No = 0, NR = 0; Item 1 (eligibility criteria and source) is not used to calculate the PEDro score; the following cut-points were used to describe the methodological quality of the studies: 9–10 (excellent), 6–8 (good), 4–5 (fair), and ≤ 3 (poor)

* = scores obtained from the PEDro database.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

BMJ Open

The Acute Effect of Breaking up Prolonged Sitting on Cognition: A Systematic Review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-050458.R2
Article Type:	Original research
Date Submitted by the Author:	11-Feb-2022
Complete List of Authors:	Chueh, Ting-Yu; National Taiwan Normal University, Physical Education and Sport Sciences Chen, Yung-Chih; National Taiwan Normal University, Physical Education and Sport Sciences Hung , Tsung-Min ; National Taiwan Normal University, Physical Education and Sport Sciences
Primary Subject Heading:	Sports and exercise medicine
Secondary Subject Heading:	Public health, Sports and exercise medicine
Keywords:	PUBLIC HEALTH, SPORTS MEDICINE, Neurophysiology < NEUROLOGY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Breaking sitting and cognition

Title: The Acute Effect of Breaking up Prolonged Sitting on Cognition: A Systematic Review

Running head: Breaks from sitting and cognition

Word count: 4612

Authors:
Ting-Yu Chueh[1], Yung-Chih Chen*[1], Tsung-Min Hung*[1, 2]

Affiliations:
[1] Department of Physical Education and Sport Sciences, National Taiwan Normal University, Taipei, Taiwan
[2] Institute for Research Excellence and Learning Sciences, National Taiwan Normal University, Taipei, Taiwan

***Correspondence:**
Yung-Chih Chen, Ph.D.
E-mail: yc.chen@ntnu.edu.tw
Department of Physical Education and Sport Sciences, National Taiwan Normal University, Taipei, Taiwan
162, Section 1, Heping E. Rd., Taipei City 106, Taiwan

Tsung-Min Hung, Ph.D., FNAK, FISSP
E-mail: ernesthungkimo@yahoo.com.tw
Department of Physical Education and Sport Sciences & Institute for Research Excellence and Learning Sciences, National Taiwan Normal University, Taipei, Taiwan
162, Section 1, Heping E. Rd., Taipei City 106, Taiwan

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Abstract

Objectives: To review the current evidence on the acute effects of interrupting prolonged periods of sitting with intermittent physical activity (PA) on cognition in healthy populations.

Design: This systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.

Methods: Studies were included if they investigated the acute effects of taking regular PA breaks from sitting on cognition in healthy populations without any cardiovascular disease, history of brain injury, or psychiatric or neurological disorder. Four electronic databases—PubMed, Scopus, MEDLINE, and ProQuest—were searched for eligible studies on September 20, 2020. Study quality was assessed using the Physiotherapy Evidence Database scale. This review is registered in PROSPERO (number: CRD42020147536).

Results: Seven studies, involving 168 participants aged between 18 and 80 years, were eligible for inclusion in this review. Three of the seven studies found positive effects of interrupting sitting with either a) 3 min of relatively high-intensity (6 km/h) walking every 30 min on attention and inhibitory control in young adults; b) hourly breaks with progressively longer durations (10–30 min) of very light-intensity cycling/walking on attention, working memory, and cognitive flexibility in adults with obesity; or c) an initial bout of continuous moderate-intensity exercise, followed by interruption of post-exercise sitting with 3-min breaks of light-intensity walking (3.2 km/h) every 30 min, on working memory in older adults with overweight.

Conclusion: Given the limited evidence with mixed findings on this topic in the literature and the heterogeneity of PA protocols across the included studies, the results regarding the effectiveness of interrupting prolonged sitting with PA breaks in improving cognition warrant further verification.

Funding: Higher Education Sprout Project by Ministry of Education. Award/Grant number is not applicable.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Key words: Sedentary behavior; Cognition; Physical activity; Breaking sitting; Brain health

For peer review only

Strengths and limitations of this study

- ✓ This review conducted a rigorous method to systematically identify the available literature.
- ✓ This review provided a quality assessment of included studies.
- ✓ The scope of this review was constrained due to the specific focus on healthy populations and acute physical activity breaks intervention.
- ✓ This review was limited to performing a meta-analysis due to the methodological heterogeneity of the included studies.
- ✓ This review only included studies published in the English language.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Introduction

Optimal cognitive performance in all cognitive domains, including attention, executive functions (EFs), and memory engagement, is important for optimizing daily functioning and reducing cognitive decline, which are critical to productivity and quality of life [1,2]. In classrooms and offices, for example, individuals require attention to focus on relevant information while relying on distinct EFs to suppress irrelevant distraction (inhibitory control), hold and mentally organize information (working memory [WM]), and adjust their behavior or thoughts based on updated demands, rules, or priorities (cognitive flexibility) to achieve goals [3-6]. These abilities have profound attribution effects on academic attainment and job performance [7,8]. In addition, a healthy cognitive state is beneficial in later life, because it reduces age-related cognitive decline [9]. Therefore, it is important to maintain and/or enhance cognition in daily life.

Sedentary behavior and physical activity (PA) are two important modifiable factors associated with cognition [10,11]. Adults typically spend the vast majority (8–10 h on average) of their waking day in sedentary behavior such as sitting and/or reclining [12,13]. Recent studies have indicated that increased sedentary behavior was associated with lower cognitive performance [14-16]. This association was partially independent of the PA level[17]. Studies also have demonstrated that prolonged sitting transiently deteriorated cognitive performance (i.e., increases problem solving and attention task errors) [18,19]. Although it has been well established that cognitive performance is enhanced after 20 to 30 min of continuous/structured exercise [20,21], individuals who engage in such exercise may be largely sedentary throughout the rest of the day [22]. Accordingly, taking regular PA breaks from sitting throughout the day may be a feasible strategy to offset the negative effects of prolonged sitting on cognition.

Taking regular PA breaks from sitting has been shown to be a practical approach to reduce the negative effects of acute prolonged sitting on cardiovascular and metabolic functions [23-25].

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Given that acute uninterrupted sitting has negative effects on cognition [18,19], related studies have extended the investigation from the physiological outcomes to the cognitive outcomes of taking regular breaks from sitting. Although recent studies have investigated the effects of taking breaks from sitting on cognition, they have not yet been subjected to a systematic review, which can provide a high level of confidence in the current findings. Furthermore, understanding the physiological mechanisms behind the acute effects of taking PA breaks from sitting on cognition is important to facilitate the design of successful PA approaches. Thus, the aims of this paper were (1) to systematically review studies investigating the acute effects of interrupting prolonged periods of sitting on cognition compared with uninterrupted sitting; (2) to provide a methodological assessment of all of the reviewed studies; and (3) to discuss the potential underlying physiological mechanisms of any effects found and provide suggestions for future studies.

Methods

This review was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines [26,27] and is registered in the International Prospective Register of Systematic Reviews (PROSPERO; identification code: CRD42020147536) [1].

Eligibility criteria

The Population, Intervention, Comparison, Outcome, and Study (PICOS) framework was used to determine the inclusion criteria for studies [26], as follows. (P) Participants: participants

[1] This study was initially planned to include studies that examined the acute effects of interrupting sitting on cerebrovascular function. However, only three studies met the eligibility criteria in the final search, and two of them involved the same dataset of cognition studies. Thus, we changed the focus of this systematic review to studies that evaluated the acute effects of interrupting sitting on cognition.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

were healthy and without any cardiovascular disease, history of brain injury, or psychiatric or neurological disorder. (I) Intervention: interventions were of an acute nature, involving interrupting prolonged periods of sitting with a regular pattern of PA at predefined intervals (e.g., every 30 min). (C) Comparator: studies included a non-treatment control group (i.e., the prolonged sitting condition). (O) Outcome: studies tested cognitive performance. (S) Study design: studies were randomized controlled or non-randomized controlled trials. Studies were excluded if they (1) involved only a single bout of PA in the intervention condition; (2) did not explicitly state the PA-related parameters, such as type, intensity, frequency, and duration; and/or (3) were not published in an English language peer-reviewed journal.

Information sources and search strategy

Four electronic databases—PubMed, Scopus, MEDLINE, and ProQuest—were searched for eligible studies from inception. The initial search was conducted on August 5, 2019, and the final search was completed on September 20, 2020. The key words used to search the titles and abstracts were discussed by the research team to maximize the chance of identifying relevant articles. The following key words were used: (*"Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting"*) *AND* (*cognitive OR cognition OR executive OR brain OR cerebral*). The full search terms used in all four databases are provided in Supplementary Appendix S1.

The inclusion and exclusion of articles were decided according to the PICOS criteria, with the screening and selection of studies being completed by two authors (TYC and YCC). First, the titles and abstracts were independently assessed by these two authors and initially coded as "yes," "no," or "maybe" for inclusion. The same two authors then reviewed the full texts of the "yes" and "maybe" studies, and disagreements regarding the inclusion of any study were resolved by discussion with a third reviewer (TMH). The reference lists of all of the included

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

articles were then searched to check for potentially relevant studies. **Figure 1** provides an overview of the selection process.

Data collection and items

The data collection was conducted by the same two authors (TYC and YCC). The authors thoroughly read the included studies and extracted the following data: (1) first author's name, publication year, and country of data collection; (2) participants' characteristics (viz., sample size, age, weight status, duration of exercise, and daily sedentary behavior); (3) study design; (4) details of the intervention protocols; and (5) outcome measures.

Synthesis methods

The results in this review were analyzed through a process of narrative synthesis. The data were first analyzed by one reviewer (TYC) and then verified by a second reviewer (YCC). As all of the necessary information could be obtained from the articles, no authors were contacted for information. A p-value of < 0.05, presented in the original studies, was used across the studies to determine the significant effects of interrupting sitting with PA on cognition. The results were converted to effect sizes (standardized mean difference) and entered into the results column of **Table 1**. Cohen's d was used to interpret the effect sizes: small ($d = 0.2$), medium ($d = 0.5$), and large ($d = 0.8$) [28].

Quality assessment

The quality of the included studies was assessed using the Physiotherapy Evidence Database (PEDro) scale [29], which produces a quantitative assessment of bias as a score out of 10, with higher scores indicating higher methodological quality. The PEDro scale assesses allocation bias (randomization, concealment of allocation), performance bias (blinding of participants and personnel), and detection bias (blinding of outcome assessment), in addition

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

to other items, including eligibility criteria, baseline comparability, retention rate, intention-to-treat, between-group statistical comparisons for at least one key outcome, point measures, and variability measures. When available, the PEDro score was obtained from the PEDro database. This was possible only for one study [30]. For studies for which the PEDro score was not available from the database, it was calculated by the two review authors (TYC and YCC) independently. Differences were resolved by discussion and adjudicated by a third author (TMH) as required.

Patient and public involvement

No patient involved.

Results

Study selection

In total, seven studies met all of the inclusion criteria and were included in this review. **Figure 1** illustrates the study selection process in a PRISMA flow diagram [26].

Study characteristics

All of the seven included studies were randomized crossover trials and comprised a total of 91 male and 77 female participants aged between 18 and 80 years. The body mass index (BMI) of the participants across all of the studies was between 18.5 and 44 kg/m^2. Most of the participants (57%) were overweight or obese (BMI > 25 kg/m^2) [30-32], and most (86%) did not report being actively engaged in regular exercise, defined as engaging in moderate to vigorous physical activity (MVPA) for at least 150 min per week. The duration of sedentary behavior across all studies was between 5 and 9 h per day. **Table 1** provides a summary of the study designs and participant characteristics.

Regarding the PA intensity during breaks from sitting, four out of the seven studies stated that the exercise was of light-intensity (walking on a treadmill at 1.6–3.6 km/h or cycling at 20

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

watts with 25–30 rpm) [30-33], two studies reported that the break consisted of moderate-intensity PA (walking on a treadmill at ~ 6 km/h) [34] and one study implemented 10 calf raises at a rate of 0.33 Hz (20 per minute) [35].

With regard to the duration and frequency of breaks, three studies implemented 3-min PA breaks from sitting every 30 min, for a total daily exercise duration of 27 [34], 30 [32], and 51 min [33]. In one study, every 60 min of sitting was interrupted with 5 min of PA, for a total daily PA duration of 30 min [36], and in another study, every 60 min of sitting was interrupted with progressively longer durations of either walking or cycling (i.e., 10, 15, 20, and 30 min), for a daily total of 150 min of very low-intensity PA [31]. One study instructed the participants to performed 10 calf raises within 30 s after every 5 or 10 min of sitting, for a total daily PA duration of 9 min (17 sets of calf exercise) [35]. The remaining study included an initial 30-min bout of moderate-intensity walking, followed by sitting interrupted every 30 min with 3 min of PA, for a total daily PA duration of 69 min [30]. **Table 1** summarizes the intervention characteristics for all of the included studies.

Results of individual studies

All of the included studies focused on EFs, three studies evaluated attention or processing speed [30,31,34], two assessed memory [31], and one study used a visual learning task [30]. Three studies found positive effects of PA breaks from sitting on EFs. In one study, compared with prolonged sitting, progressively longer durations of both light-intensity walking and cycling (i.e., 10, 15, 20, and 30 min) after every 60 min of sitting resulted in greater attention, WM, and cognitive flexibility, with small to large effect sizes (d = 0.62–0.91) [31]. Interestingly, cycling resulted in a better improvement in attention than walking, with a medium effect size (d = 0.63). Taking 3-min breaks of moderate-intensity PA (6.0 km/h, rating of perceived exertion [RPE] = 12–14) every 30 min resulted in improved inhibitory control with a large effect size (d = 0.74) [34], whereas taking 3-min breaks of light-intensity walking (3.2 km/h)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

every 30 min [32,33] or 5-min breaks of moderate-intensity walking (RPE = 12–13) every 60 min showed no significant effect on any cognitive performance measure [36]. Three-minute breaks of light-intensity PA (3.2 km/h) every 30 min after an initial bout of continuous moderate-intensity exercise (65–75% of the age-predicted maximal heart rate) had a beneficial effect on WM (d = 0.25) [30]. One study found adverse effects of intermittent 10 calf raise exercises on inhibitory control (d = -0.85) [35]. Two studies reported the positive effects of taking 3-min breaks of moderate-intensity PA every 30 min or progressively longer breaks of light-intensity walking and cycling every 60 min on attention, with medium to large effect sizes (d = 0.42 to 1.1) [31,34]. Interestingly, cycling resulted in a better improvement in attention than walking, with a medium effect size (d = 0.63). No significant effects of interrupting sitting with PA was found in other cognitive domains.

Methodological quality

The included studies had a mean PEDro score of 4.1, indicating "fair" methodological quality. All studies reported to have performed random allocation of participants to the trials, but only one study reported performing concealed allocation [30]. This suggests the presence of allocation biases in the other studies. None of the studies mentioned the blinding of participants and/or researchers because it is impossible to blind an intervention that requires the participants to perform activities. All of the studies provided a key outcome measure for more than 85% of their participants, and all of the studies adopted a crossover trial design, which meant that all participants received all of the designed treatments. Nevertheless, none of the studies explicitly stated whether at least one main outcome was included in the statistical analysis (intention-to-treat analysis) of all of the participants, regardless of any subsequent withdrawal from treatment. This suggests the presence of biased comparisons between the treatment arms. Finally, all of the studies provided statistical comparisons and valid measures for at least one key outcome measure of interest. Detailed results of the PEDro assessment are presented in **Table 2**.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Discussion

The purpose of this study was to review the results of studies that investigated whether interrupting prolonged periods of sitting with PA breaks would improve cognition. Seven studies were identified that investigated the acute effects of such interruptions in healthy individuals. All of these studies were published recently (between 2016 and 2020). Three of the seven studies suggested that taking regular PA breaks from sitting offers short-term benefits for cognition in healthy adults. However, no definite conclusions could be drawn regarding the effectiveness of taking PA breaks from sitting in improving cognition due to the heterogeneity of the PA protocols across the included studies and the small number of available studies.

The acute effects of taking PA breaks from sitting on cognition

This review suggests that taking PA breaks from sitting, specifically their intensity and/or volume, might play a potential role in regulation of cognitive performance. For example, compared with prolonged sitting, taking 3-min breaks of moderate-intensity walking (6 km/h) every 30 min was reported to result in better attention and inhibitory control in young female adults [34]. However, the same protocol (3 min of PA every 30 min) but with light-intensity walking (3.2 km/h) produced no effect on episodic memory, inhibitory control, or WM in overweight/obese middle-aged adults [32] or on attention or WM in young adults [33]. Studies have reported that exercise intensity may be an important determinant of cognitive benefits [37,38]. A meta-analysis demonstrated that the positive acute effect of exercise on cognition is seen when the exercise intensity is moderate, but the effect diminishes when the exercise intensity is very light [39]. One possible explanation is that greater exercise intensity is needed to induce sufficient levels of norepinephrine (NE) to facilitate cognitive performance [40]. Thus, the intensity of PA during breaks from sitting may be one of the factors determining the effectiveness of the breaks.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

However, taking higher-intensity PA breaks is not always beneficial if the frequency of such breaks is low. For example, another study of non-obese adults investigated the effects of a similar walking speed (~5.7 km/h), but less frequent (hourly) and slightly longer (5 min), walking breaks from sitting and found no effect on the measures of inhibitory control or cognitive flexibility [36]. This implies that the benefits of taking relatively high intensity PA breaks from sitting might be reduced by having infrequent breaks. Interestingly, taking hourly low-intensity PA breaks might still benefit cognition if these breaks are sufficiently long (i.e., 10, 15, 20, and 30 min). As indicated in one study [31], both very low-intensity cycling (work rate = 20 W, cadence = 25–30 rpm) and walking (1.6 km/h) resulted in better attention (the detection test), cognitive flexibility (the set-shifting test), and WM (the one-back test) compared with prolonged sitting. Despite the low intensity of PA breaks, the longer duration of PA breaks adopted in this study, which resulted in a greater amount of exercise (150 min in total), might have had a compensating effect. This finding implies that volume of exercise should be considered and carries important implication for some individuals who are unable to engage in moderate to vigorous intensity exercise. Collectively, these results suggest that the effects of taking breaks from sitting on cognitive function are dependent on the intensity, frequency, and duration of the PA performed during the breaks. Future studies are recommended to determine the minimum number and duration of low-intensity PA breaks from sitting required for a meaningful improvement in cognition.

The study that evaluated the effects of a morning bout of 30 min of moderate-intensity exercise with or without subsequent light-intensity PA breaks every 30 min on cognitive performance in obese/overweight older adults provides another fruitful line of research [30]. A single bout of moderate-intensity PA before engaging in prolonged sitting showed better EFs, but not WM, relative to the prolonged sitting condition, whereas the interruption of post-exercise sitting with several additional bouts of light-intensity PA led to improved WM, but not EFs [30]. These results suggest that a single bout of continuous moderate-intensity exercise

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

with or without subsequent PA breaks provide benefits in different cognitive domains. These divergent effects may be attributed to individual differences in the level of aerobic fitness and the timing and order of the administration of cognitive tests following the cessation of exercise. It is reasonable to assume that the participants in that study had lower aerobic fitness, based on the provided demographic data (old age, sedentary lifestyle, and obese/overweight status). In this study, the test of EFs was administered almost immediately (approximately 2 min) after the 3-min light-intensity PA breaks, followed by the WM test and other cognitive assessments. A previous meta-analytic review indicated that the largest effect of lower-intensity PA on cognitive performance was immediately after the PA, whereas that of higher-intensity activities was after a delay of ~10 min [39]. However, an individual's level of aerobic fitness may moderate the physiological demands and/or perception of the physical activity stimulus [41]. Chang, et al. [39] observed that aerobic fitness was a moderating variable when cognition was assessed immediately after the cessation of a bout of PA. Thus, the beneficial effects of a single bout of continuous moderate-intensity exercise with subsequent light-intensity PA breaks on EFs in this study may have been diminished due to the lower aerobic fitness level of the participants and the small time interval between exercise cessation and EF assessment. Moreover, when the aim is to maintain WM, it appears that it is necessary to avoid prolonged sitting even after an exercise session.

Of note, one study found adverse effects of frequent calf raise exercises on inhibitory control in young adults [35]. The impaired inhibitory control performance was accompanied by a decreased prefrontal cortex perfusion, as measured by near-infrared spectroscopy, supporting that prefrontal cortex activity could be a potential mechanism for modulating cognitive performance [6]. The negative effect of acute exercise on EFs was observed only after a bout of high-intensity exercise [39]. We found no study reporting a negative effect on cognition following a short bout of low-intensity exercise, such as calf raises. The reason for this result is unknown; therefore, future studies should attempt to replicate this result and investigate the reason.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Taken together, the literature reviewed suggests that to understand the effects of PA breaks from prolonged sitting on cognitive performance, the effects of the frequency, intensity, and duration of the PA undertaken during the breaks need to be considered concurrently. The current evidence suggests that a) short, frequent breaks (at least every 30 min) involving moderate-intensity PA (walking at least 6 km/h); b) longer, hourly breaks involving even very low-intensity PA; and c) a single bout of 30-min continuous moderate-intensity exercise with subsequent short PA breaks are all beneficial to cognitive performance (e.g., attention, EFs, or memory-related tasks). However, the heterogeneity of PA protocols across the reviewed studies and the small number of available studies make direct comparisons of study results difficult. Furthermore, the findings of the effectiveness of PA breaks from sitting in improving cognition need to be further confirmed, because fewer than half of the studies included reported significant positive results.

Potential mechanisms underlying the effects of interrupting sitting on cognition

Despite the inconsistent results across the reviewed studies, the available evidence suggests that taking PA breaks from prolonged sitting exerts acute beneficial effects on cognition. Several potential mechanisms may explain why interrupting periods of sitting is beneficial to brain function.

First, research in both humans and animal models suggests that the activation of the locus coeruleus and the associated release of NE play important roles in influencing the attention state, which may modulate cognitive performance [42,43]. NE release is at least partly modulated by the amount of PA. Studies have found that prolonged uninterrupted sitting is associated with increased levels of fatigue [44] and decreased plasma levels of NE [32], whereas moderate-intensity exercise can induce NE release (indexed by salivary alpha-amylase concentrations) and enhance cognitive performance [32,45]. Two studies that used either frequent breaks of relatively light-intensity PA (every 30 min) or less frequent breaks of moderate-intensity PA (every 60

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

min) found no beneficial effects on the plasma and urinary levels of NE or cognitive performance [32,36]. Moderate-intensity PA breaks every 30 min; progressively increasing the duration of breaks, even at light-intensity PA; and performing an initial bout of continuous moderate-intensity exercise for 30 min before subsequent light-intensity PA breaks during sitting, all resulted in improved cognitive performance compared with the prolonged sitting condition [31,34]. It is known that cognitive performance is enhanced with increasing levels of NE [40], which, in turn, is affected by PA. Thus, individuals interested in taking regular physical activity breaks from sitting for cognitive benefits should take into account the combination of intensity, frequency, and duration.

Secondly, glucose plays a vital role in supplying energy to cranial nerves and so glucose metabolism may influence cognition. The glucose-centric hypothesis postulates that exposure to acute hyperglycemia impairs cognition [11]. Indeed, young and middle-aged people with diabetes have shown impaired performance on measures of memory, visual perception, and attention. This may be partly due to brain injury through neurodegeneration, frontal and hippocampal atrophy, and injury to the white matter microstructure [46]. Studies have demonstrated that acute glucose ingestion (i.e., blood glucose concentrations increasing from 90 to 142 mg/dl) reduces the global cerebral metabolic rate of oxygen [47] and regional cerebral blood flow (CBF) [48], which result in poorer cognitive performance [49]. Regularly interrupting periods of sitting has been shown to reduce abrupt elevations in postprandial glucose concentrations [23,25]. One study found that in healthy, normal-weight adults, taking short but frequent PA breaks (every 30 min) resulted in higher CBF than taking longer but less frequent PA breaks (every 120 min)[50]. This is in line with the results of a previous study that examined the acute effects of taking regular PA breaks with different bouts of PA on blood glucose [51]. These findings suggest that regularly taking PA breaks after meals might be beneficial to cognition by preventing both the reductions in cerebral oxygen and regional CBF and repeated subtle brain injuries. On the other hand, it has been shown that hypoglycemia (from 100 to 48

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

mg/dl) also acutely reduces blood oxygen level-dependent cortical activation and cognitive function [52], as the brain requires a high level of energy supply [53]. Therefore, to avoid hyperinsulinemic hypoglycemia after meal consumption, taking breaks from sitting may be a useful means of maintaining blood glucose concentration within a stable range. In summary, studies suggest that taking breaks from sitting might benefit cognition by means of affecting glucose metabolism.

Limitations and suggestions for future research

Some caution is warranted when interpreting the findings of this review. First, the results cannot be generalised beyond the specific populations studied, i.e. children, adolescents, and individuals with chronic disease such as type 2 diabetes. The current literature has not focused on school-aged children. School-aged children spend the majority of class time sitting and this might negatively affect their cognitive function (i.e., attention state) [54]. In addition, individuals with type 2 diabetes showed impaired glucose metabolism that might, according to the glucose-centric hypothesis, modulate the relationship between PA and cerebrovascular function [11]. Thus, these populations need to be independently examined. Second, given the heterogeneity of PA protocols, cognitive assessments, and sample size across the included studies, more studies with larger samples are needed to clarify the role of different PA characteristics (i.e., frequency, intensity, duration, and type) in the effect of PA breaks from sitting on distinct cognitive domains so as to provide more precise recommendations for public health. Third, this review exclusively focused on studies that assessed the acute effects of PA breaks on cognition. Future research should explore the accumulative effects of multiple days of sitting with or without PA breaks on cognition. Fourth, there is a need to further explore the underlying mechanisms using biochemical and physiological measures, such as neural activity in the locus coeruleus NE system and glucose metabolism. Finally, future studies should examine the isolated

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

contributions of interrupting prolonged sitting versus continuous PA to improvements in cognition.

Conclusions

The current review provides preliminary evidence that the intensity and/or volume of PA breaks from prolonged sitting may determine the effects of such breaks on cognition. However, due to the heterogeneity of PA protocols across studies and limited evidence, firm conclusions regarding the effectiveness of interrupting sitting in improving cognition cannot be drawn. Further studies are warranted to examine the duration, frequency, and intensity of PA breaks required to optimize cognitive performance.

Conflict of interest: There is no conflict of interest to report.

Ethics approval: Ethical approval was not applicable because this study retrieved data from already published studies.

Data availability statement: No additional data available.

Author contributions: TYC conceived the study concept; contributed to the study design, literature search, and data screening and extraction; assessed the article quality; and wrote the first draft of the manuscript. YCC conceived the study concept; contributed to the study design, literature search, and data screening and extraction; assessed the article quality; and critically reviewed and revised the manuscript. TMH conceived the study concept, contributed to the study design, assisted with data screening and extraction and quality

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

assessment, and critically reviewed and revised the manuscript. All of the authors approved the submission.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

References

1 Hill, N. L. *et al.* Subjective cognitive impairment and quality of life: a systematic review. *International Psychogeriatrics* **29**, 1965-1977 (2017).

2 Sanz, J. H. *et al.* Executive function and psychosocial quality of life in school age children with congenital heart disease. *J Pediatr* **202**, 63-69 (2018).

3 Diamond, A. Executive functions. *Annu Rev Psychol* **64**, 135-168 (2013).

4 Miyake, A. *et al.* The unity and diversity of executive functions and their contributions to complex "frontal lobe" tasks: A latent variable analysis. *Cognitive psychology* **41**, 49-100 (2000).

5 Zelazo, P. D. Executive function: Reflection, iterative reprocessing, complexity, and the developing brain. *Developmental Review* **38**, 55-68 (2015).

6 Miller, E. K. & Cohen, J. D. An integrative theory of prefrontal cortex function. *Annual review of neuroscience* **24**, 167-202 (2001).

7 Lang, J. W., Kersting, M., Hülsheger, U. R. & Lang, J. General mental ability, narrower cognitive abilities, and job performance: The perspective of the nested-factors model of cognitive abilities. *Personnel psychology* **63**, 595-640 (2010).

8 Rohde, T. E. & Thompson, L. A. Predicting academic achievement with cognitive ability. *Intelligence* **35**, 83-92 (2007).

9 Buckner, R. L. Memory and executive function in aging and AD: multiple factors that cause decline and reserve factors that compensate. *Neuron* **44**, 195-208 (2004).

10 Erickson, K. I. *et al.* Physical Activity, Cognition, and Brain Outcomes: A Review of the 2018 Physical Activity Guidelines. *Medicine and science in sports and exercise* **51**, 1242-1251 (2019).

11 Wheeler, M. J. *et al.* Sedentary behavior as a risk factor for cognitive decline? A focus on the influence of glycemic control in brain health. *Alzheimer's & Dementia: Translational Research & Clinical Interventions* **3**, 291-300 (2017).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12 Castro, O., Bennie, J., Vergeer, I., Bosselut, G. & Biddle, S. J. How sedentary are university students? A systematic review and meta-analysis. *Prevention Science* **21**, 332-343 (2020).

13 Van Dyck, D. *et al.* International study of objectively measured physical activity and sedentary time with body mass index and obesity: IPEN adult study. *International Journal of Obesity* **39**, 199-207 (2015).

14 Bakrania, K. *et al.* Associations between sedentary behaviors and cognitive function: Cross-sectional and prospective findings from the UK biobank. *American journal of epidemiology* **187**, 441-454 (2017).

15 Falck, R. S., Davis, J. C. & Liu-Ambrose, T. What is the association between sedentary behaviour and cognitive function? A systematic review. *Br J Sports Med* **51**, 800-811 (2017).

16 Vancampfort, D. *et al.* Mild cognitive impairment and sedentary behavior: a multinational study. *Experimental gerontology* **108**, 174-180 (2018).

17 Ku, P.-W., Liu, Y.-T., Lo, M.-K., Chen, L.-J. & Stubbs, B. Higher levels of objectively measured sedentary behavior is associated with worse cognitive ability: Two-year follow-up study in community-dwelling older adults. *Experimental gerontology* **99**, 110-114 (2017).

18 Triglav, J. *et al.* Physiological and cognitive measures during prolonged sitting: Comparisons between a standard and multi-axial office chair. *Applied ergonomics* **78**, 176-183 (2019).

19 Baker, R., Coenen, P., Howie, E., Williamson, A. & Straker, L. The short term musculoskeletal and cognitive effects of prolonged sitting during office computer work. *International journal of environmental research and public health* **15**, 1678 (2018).

For peer review only

20 Oberste, M. *et al.* Effects and moderators of acute aerobic exercise on subsequent interference control: a systematic review and meta-analysis Running title: Acute exercise and interference control. *Frontiers in Psychology* **10**, 2616 (2019).

21 Ludyga, S., Gerber, M., Brand, S., Holsboer-Trachsler, E. & Pühse, U. Acute effects of moderate aerobic exercise on specific aspects of executive function in different age and fitness groups: A meta-analysis. *Psychophysiology* **53**, 1611-1626 (2016).

22 Dempsey, P. C., Owen, N., Biddle, S. J. & Dunstan, D. W. Managing sedentary behavior to reduce the risk of diabetes and cardiovascular disease. *Current diabetes reports* **14**, 522 (2014).

23 Chen, Y.-C., Betts, J. A., Walhin, J.-P. & Thompson, D. Adipose Tissue Responses to Breaking Sitting in Men and Women with Central Adiposity. *Medicine and science in sports and exercise* **50**, 2049-2057 (2018).

24 Dunstan, D. W. *et al.* Breaking up prolonged sitting reduces postprandial glucose and insulin responses. *Diabetes care* **35**, 976-983 (2012).

25 Saunders, T. J. *et al.* The acute metabolic and vascular impact of interrupting prolonged sitting: a systematic review and meta-analysis. *Sports medicine* **48**, 2347-2366 (2018).

26 Moher, D. *et al.* Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic reviews* **4**, 1 (2015).

27 Moher, D., Liberati, A., Tetzlaff, J. & Altman, D. G. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Int J Surg* **8**, 336-341 (2010).

28 Cohen, J. Statistical Power Analysis for the Behavioral Sciences–Second Edition. 12 Lawrence Erlbaum Associates Inc. *Hillsdale, New Jersey* **13** (1988).

29 Maher, C. G., Sherrington, C., Herbert, R. D., Moseley, A. M. & Elkins, M. Reliability of the PEDro scale for rating quality of randomized controlled trials. *Physical therapy* **83**, 713-721 (2003).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

30 Wheeler, M. J. *et al.* Distinct effects of acute exercise and breaks in sitting on working memory and executive function in older adults: a three-arm, randomised cross-over trial to evaluate the effects of exercise with and without breaks in sitting on cognition. *Br J Sports Med* **54**, 776-781 (2020).

31 Mullane, S. L., Buman, M. P., Zeigler, Z. S., Crespo, N. C. & Gaesser, G. A. Acute effects on cognitive performance following bouts of standing and light-intensity physical activity in a simulated workplace environment. *J Sci Med Sport* **20**, 489-493 (2017).

32 Wennberg, P. *et al.* Acute effects of breaking up prolonged sitting on fatigue and cognition: a pilot study. *BMJ open* **6** (2016).

33 Vincent, G. E. *et al.* The impact of breaking up prolonged sitting on glucose metabolism and cognitive function when sleep is restricted. *Neurobiology of sleep and circadian rhythms* **4**, 17-23 (2018).

34 Chrismas, B. C., Taylor, L., Cherif, A., Sayegh, S. & Bailey, D. P. Breaking up prolonged sitting with moderate-intensity walking improves attention and executive function in Qatari females. *PloS one* **14**, e0219565 (2019).

35 Stoner, L. *et al.* Effects of acute prolonged sitting on cerebral perfusion and executive function in young adults: A randomized cross-over trial. *Psychophysiology* **56**, e13457 (2019).

36 Bergouignan, A. *et al.* Effect of frequent interruptions of prolonged sitting on self-perceived levels of energy, mood, food cravings and cognitive function. *International Journal of Behavioral Nutrition and Physical Activity* **13**, 1-12 (2016).

37 Bailey, B. W. *et al.* The impact of exercise intensity on neurophysiological indices of food-related inhibitory control and cognitive control: A randomized crossover event-related potential (ERP) study. *NeuroImage* **237**, 118162 (2021).

For peer review only

38 Zimmer, P. *et al.* The effects of different aerobic exercise intensities on serum serotonin concentrations and their association with Stroop task performance: a randomized controlled trial. *European journal of applied physiology* **116**, 2025-2034 (2016).

39 Chang, Y., Labban, J., Gapin, J. & Etnier, J. The effects of acute exercise on cognitive performance: a meta-analysis. *Brain research* **1453**, 87-101 (2012).

40 Berridge, C. W. & Spencer, R. C. Differential cognitive actions of norepinephrine a2 and a1 receptor signaling in the prefrontal cortex. *Brain research* **1641**, 189-196 (2016).

41 American College of Sports Medicine. *ACSM's guidelines for exercise testing and prescription (10th ed.)*. (Lippincott williams & wilkins, 2018).

42 Aston-Jones, G. & Cohen, J. D. An integrative theory of locus coeruleus-norepinephrine function: adaptive gain and optimal performance. *Annu. Rev. Neurosci.* **28**, 403-450 (2005).

43 Gelbard-Sagiv, H., Magidov, E., Sharon, H., Hendler, T. & Nir, Y. Noradrenaline modulates visual perception and late visually evoked activity. *Current Biology* **28**, 2239-2249. e2236 (2018).

44 Dempsey, P. C. *et al.* Prolonged uninterrupted sitting increases fatigue in type 2 diabetes. *Diabetes research and clinical practice* **135**, 128-133 (2018).

45 Segal, S. K., Cotman, C. W. & Cahill, L. F. Exercise-induced noradrenergic activation enhances memory consolidation in both normal aging and patients with amnestic mild cognitive impairment. *Journal of Alzheimer's disease* **32**, 1011-1018 (2012).

46 Weinstein, G. *et al.* Glucose indices are associated with cognitive and structural brain measures in young adults. *Neurology* **84**, 2329-2337 (2015).

47 Xu, F. *et al.* Acute effect of glucose on cerebral blood flow, blood oxygenation, and oxidative metabolism. *Human brain mapping* **36**, 707-716 (2015).

48 Page, K. A. *et al.* Effects of fructose vs glucose on regional cerebral blood flow in brain regions involved with appetite and reward pathways. *Jama* **309**, 63-70 (2013).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

49 Marshall, R. S. *et al.* Recovery of brain function during induced cerebral hypoperfusion. *Brain* **124**, 1208-1217 (2001).

50 Carter, S. E. *et al.* Regular walking breaks prevent the decline in cerebral blood flow associated with prolonged sitting. *Journal of Applied Physiology* **125**, 790-798 (2018).

51 Peddie, M. C. *et al.* Breaking prolonged sitting reduces postprandial glycemia in healthy, normal-weight adults: a randomized crossover trial. *The American journal of clinical nutrition* **98**, 358-366 (2013).

52 Rosenthal, J. M. *et al.* The effect of acute hypoglycemia on brain function and activation: a functional magnetic resonance imaging study. *Diabetes* **50**, 1618-1626 (2001).

53 Rolfe, D. & Brown, G. C. Cellular energy utilization and molecular origin of standard metabolic rate in mammals. *Physiological reviews* **77**, 731-758 (1997).

54 Pontifex, M. B., Parks, A. C., Henning, D. A. & Kamijo, K. Single bouts of exercise selectively sustain attentional processes. *Psychophysiology* **52**, 618-625 (2015).

For peer review only

Breaking sitting and cognition

Table 1 Characteristics of the included studies

Author	Participants	Design	Arms	Outcomes	Results
Wheeler et al. (2020) /Australia	N = 67 (female = 35); Age: ≥ 55 to ≤ 80 years (M ± SD: 67 ± 7 years); Weight status: overweight/obese (BMI: ≥ 25 to < 45 kg/m^2); MVPA < 150 min/week; sitting time > 5 h/day	Randomized cross-over	Trial started: ~8:00; #**1. SIT:** sitting (8 h); **2. EX+SIT:** sitting (1 h), continuous moderate-intensity (65–75% of the age-predicted HR_{max}) walking (30 min), sitting (6.5 h); total PA = 30 min #**3. EX+BR:** sitting (1 h), continuous moderate-intensity (65–75% of the age-predicted HR_{max}) walking (30 min), sitting interrupted every 30 min with 3 min of light-intensity walking (3.2 km/h), sitting (6.5 h); total PA = 69 min.	1. Psychomotor function and processing speed (the detection test) 2. Attention (the identification test); 3.Visual learning (the one card learning test) 4. EFs (the Groton Maze Learning Test) 5.EF-WM (one-back and two-back tests)	**Psychomotor function, attention, visual learning, and EFs:** EX + BR = SIT **EF-WM:** EX+BR > SIT (ES = 0.25)
Stoner et al. (2019)/Hill	N = 20 (female = 14); Age: 18–35 years (M ± SD: 22 ± 3 years); Weight status: normal/overweight (BMI = 25.5 ± 6.1 kg/m^2); MVPA < 150 min/week; sitting time (M ± SD: 8.2 ± 2.2 h/day)	Randomized cross-over	Trial started: 6:00–10:00; #**1. SIT:** sitting (3 h) #**2. BR:** sitting interrupted with 10 calf raises at the 15- and 20-min points, and then every 10 min during the following sitting time; total PA = ~9 min.	EF-IC (the Stroop test)	**EF-IC:** BR < SIT (ES = –0.85)
Chrismas et al. (2019)/UK	N = 17 (female = 17); Age: 21–24 years (M = NR); Weight status and MVPA: NR; sitting time > 7 h/day	Randomized cross-over	Trial started: ~8:00; #**1. SIT:** sitting (5 h) #**2. BR:** sitting interrupted every 30 min with 3 min of moderate-intensity walking (6.0 km/h, RPE = 12–14); total PA = 27 min.	1. Attention and vigilance (rapid visual information processing and choice reaction time) 2. EF-IC (the Stroop test) 3. EF-WM (serial 3's and 7's subtraction tasks)	**Attention and vigilance:** BR > SIT (ES = 0.89) **EF-IC**; BR > SIT (ES = 0.75) **EF-WM**: BR = SIT
Vincent et al (2018) /Australia	N = 6 (female = 0); Age: 20–35 years (M ± SD: 27 ± 4 years); Weight status: lean (waist circumference = ~84 ± 7 cm); MVPA and sitting time: NR	Randomized cross-over	#**1. SIT:** sitting (7 h) #**2. BR:** sitting interrupted every 30 min with 3 min of light-intensity walking (3.2 km/h); total PA = 51 min.	EF-WM (the digit symbol substitution test)	**No significant effects**

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Mullane, Buman, Zeigler, Crespo and Gaesser (2017)/USA	N = 9 (female = 7); Age: 18–58 years (M ± SD: 30 ± 15 years); Weight status: overweight/obese (BMI = 29 ± 3 kg/m^2); MVPA < 150 min/week; sitting time: NR	Randomized cross-over	Trial started: 8:00; #**1. SIT:** sitting (8 h) **2. STAND:** standing #**3. CYCLE:** cycling (work rate = 20 W, cadence = 25–30 rpm) #**4. WALK:** walking (1.6 km/h); STAND, CYCLE, and WALK for a predetermined time hourly: 10 min at 08:50 and 09:50, 15 min at 10:45 and 11:45, 20 min at 12:40 and 13:20, and 30 min at 14:00 and 15:30; total PA = 150 min.	1. Psychomotor function (the detection test) 2. EF-WM (the one-back test) 3. EFs (the set-shifting test)	**Psychomotor function:** CYCLE > SIT (ES = 1.1); WALK > SIT (ES = 0.42); CYCLE > WALK (ES = 0.63) **EF-WM:** CYCLE > SIT (ES = 0.91);WALK > SIT (ES = 0.74) **EF-CF:** CYCLE > SIT (ES = 0.78); WALK > SIT (ES = 0.66)
Bergouignan et al. (2016)/USA	N = 30 (female = 9); Age: 25–50 years (M ± SD: 31 ± 6 years) ; Weight status: non-obese (BMI = 18.5–29.9 kg/m^2); MVPA < 150 min/week; sitting time > 9 h/day	Randomized cross-over	Trial started: 8:00; #**1. SIT:** sitting (6 h) **2. EX:** continuous moderate-intensity (RPE = 12–13) walking (30 min); total PA = 30 min #**3. BR:** sitting interrupted every 1 h with 5 min of moderate-intensity walking (RPE = 12–13); total PA = 30 min.	EFs (Flanker task and trail-making test)	**No significant effects**
Wennberg et al. (2016)/Australia	N = 19 (female = 9); Age: 45–75 years (M ± SD: 60 ± 8 years); Weight status: overweight/obese (BMI: ≥ 25 to ≤ 45 kg/m^2); MVPA < 150 min/week; sitting time > 5 h/day	Randomized cross-over	Trial started: 7:00–8:00; #**1. SIT:** sitting (7 h) #**2. BR:** sitting interrupted every 30 min with 3 min of light-intensity walking (3.2 km/h, RPE = 9); total PA = 30 min.	1. EFs (the Flanker task and Stroop test) 2. Episodic memory (a face-name association test) 3. EF-WM (the n-back task and letter memory test)	**No significant effects**

Abbreviations: M = mean; MVPA = moderate-to-vigorous intensity physical activity; SIT = sitting; EX = continuous exercise; BR = breaking sitting; HR_{max} = maximal heart rate; PA = physical activity; BMI = body mass index; CYCLE = cycling; NR = not reported; RPE = Rating of perceived exertion (Borg scale); EF = executive function; IC = inhibitory control; WM = working memory; # = included arms for review; In the result column: > = positive effect of interrupting sitting ($p < 0.05$); < = negative effect of interrupting sitting ($p < 0.05$); ES = effect size (standardized mean difference)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Breaking sitting and cognition

Table 2 Quality assessment of the included studies

Study	Eligibility criteria and source	Random allocation	Concealed allocation	Baseline comparability	Blinding of subjects	Blinding of therapists	Blinding of assessors	Completeness of follow-up	Intention -to-treat	Between-group comparison	Point and variability measures	Total scores
Wheeler et al. (2020)*	Yes	Yes	Yes	No	No	No	No	Yes	No	Yes	Yes	5
Stoner et al. (2019)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Chrismas et al. (2019)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Mullane et al. (2017)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Vincent et al. (2018)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Bergouignan et al. (2016)	Yes	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Wennberg et al. (2016)	No	Yes	No	No	No	No	No	Yes	No	Yes	Yes	4
Mean score (SD)	4.1 (0.4)											

PEDro scoring: Yes = 1, No = 0, NR = 0; Item 1 (eligibility criteria and source) is not used to calculate the PEDro score; the following cut-points were used to describe the methodological quality of the studies: 9–10 (excellent), 6–8 (good), 4–5 (fair), and ≤ 3 (poor)

* = scores obtained from the PEDro database.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

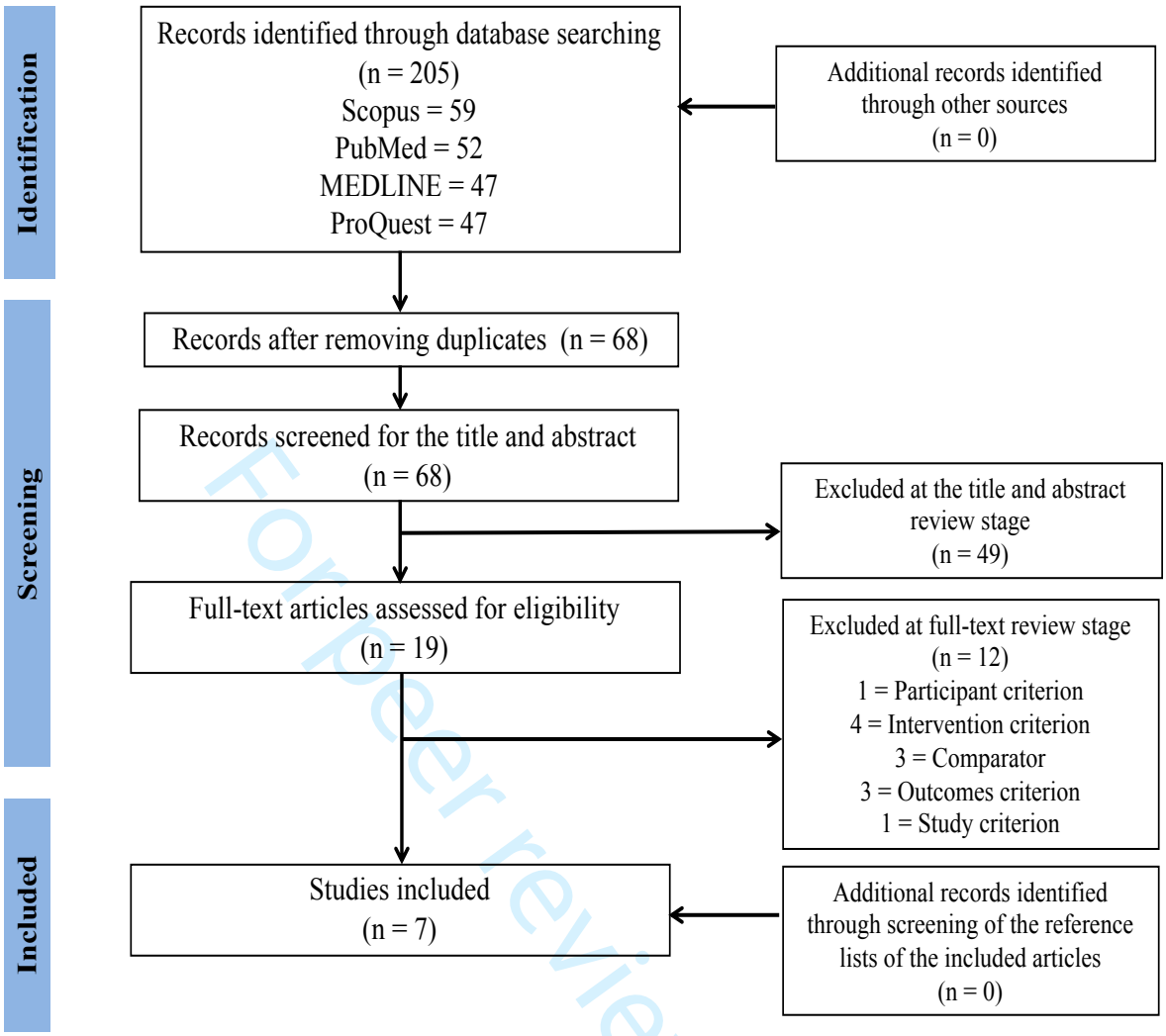


Figure 1. Flow diagram of the selection process showing the identification, screening, eligibility, and inclusion of studies.

For peer review only

Supplementary Appendix S1

Scopus	PubMed	MEDLINE	ProQuest
(TITLE-ABS ("Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting") AND TITLE-ABS (cognitive OR cognition OR executive OR brain OR cerebral))	("Prolonged sitting" [Title/Abstract] OR "Interrupted sitting" [Title/Abstract] OR "breaking sitting" [Title/Abstract] OR "uninterrupted sitting" [Title/Abstract] OR "continuous sitting"[Title/Abstract]) AND (cognitive [Title/Abstract] OR cognition [Title/Abstract] OR executive [Title/Abstract] OR brain [Title/Abstract] OR cerebral[Title/Abstract])	(TI ("Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting") AND TI (cognitive OR cognition OR executive OR brain OR cerebral)) OR (AB ("Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting") AND AB (cognitive OR cognition OR executive OR brain OR cerebral))	(ti("Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting") AND ti(cognitive OR cognition OR executive OR brain OR cerebral)) OR (ab("Prolonged sitting" OR "Interrupted sitting" OR "breaking sitting" OR "uninterrupted sitting" OR "continuous sitting") AND ab(cognitive OR cognition OR executive OR brain OR cerebral))

For peer review only



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	See title page
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	p.2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	p.1 to 2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	p.2
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	p.2
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	p.3
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	p.3
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	p.3
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	p.5
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	p.5
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	p.5
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	p.5
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NR
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	p.5
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	p.5
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	p.5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	NR
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NR
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NR
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NR

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NR
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	p.6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	p.4
Study characteristics	17	Cite each included study and present its characteristics.	p.6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	p.8
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	p.7
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	NR
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NR
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NR
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NR
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	NR
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NR
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	p.8 to 9
	23b	Discuss any limitations of the evidence included in the review.	p.14
	23c	Discuss any limitations of the review processes used.	p.14
	23d	Discuss implications of the results for practice, policy, and future research.	p.14
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	p.2
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	p.2
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	p.2
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	p.15
Competing interests	26	Declare any competing interests of review authors.	p.15
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	p.3

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2020 Checklist

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

For peer review only